

T3 UV one-loop RGE comparison

Each table is written in the conventional form $16\pi^2 \mu dX/d\mu = \beta_X^{(1)}$. A dash means that the coupling is not registered for that representation assignment.

Conventions and the RGBeta gauge-coupling conversion

For ordinary couplings the report uses the quantity returned by RGBeta directly. For gauge couplings, RGBeta uses g^2 as its running variable, so its one-loop output is proportional to $d(g^2)/d\ln\mu$. By the chain rule,

$$\frac{d(g^2)}{d\ln\mu} = 2g \frac{dg}{d\ln\mu} = 2g \mu \frac{dg}{d\mu}.$$

Therefore, to present the gauge RGE in the same convention as every other table, $16\pi^2 \mu dg/d\mu = \beta_g^{(1)}$, the RGBeta gauge result is divided by $2g$. This is only a change of running variable; no physics or loop factor is being altered.

Notation

S_1 and S_2 denote the two BSM scalar multiplets and F the BSM fermion multiplet. The columns d_X give the $SU(2)_L$ representation dimensions and Y_X the hypercharges. A dagger denotes Hermitian conjugation, T transpose, $*$ complex conjugation, and Tr a trace over flavour indices.

Coupling glossary

The interaction terms below are schematic: gauge-index contractions and conjugations are fixed by each model’s quantum numbers. For representation-dependent quartics, labels such as Adj and Cross distinguish independent $SU(2)_L$ invariant contractions.

Symbol	Meaning	Schematic Lagrangian term	Notes
g_Y	$U(1)_Y$ gauge coupling	Contained in $D_\mu = \partial_\mu + ig_Y Y B_\mu + \dots$	Hypercharge gauge interaction.
g_2	$SU(2)_L$ gauge coupling	Contained in $D_\mu = \partial_\mu + ig_2 T^a W_\mu^a + \dots$	Strongly sensitive to the $SU(2)_L$ representations of the new fields.
g_3	$SU(3)_c$ gauge coupling	Contained in the QCD covariant derivative.	The T3 fields used here are colour singlets, so their one-loop contribution to β_{g_3} vanishes.
Y_u	SM up-type Yukawa matrix	$-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$	SM Yukawa interaction.
Y_d	SM down-type Yukawa matrix	$-\bar{Q} Y_d H d + \text{h.c.}$	SM Yukawa interaction.
Y_e	SM charged-lepton Yukawa matrix	$-\bar{L} Y_e H e + \text{h.c.}$	SM charged-lepton Yukawa interaction.
y_1	T3 Yukawa matrix	$-\bar{L} y_1 F S_1^{(\dagger)} + \text{h.c.}$	Schematic only: the required conjugation and $SU(2)$ invariant tensor depend on the T3 representation assignment.
y_2	T3 Yukawa matrix	$-\bar{L} y_2 F S_2^{(\dagger)} + \text{h.c.}$	Schematic only. Together with y_1 and λ_{T3} it participates in the neutrino-mass loop.

Symbol	Meaning	Schematic Lagrangian term	Notes
M_F	BSM fermion mass matrix	$-\bar{F} M_F F$ (Dirac/vector-like case), or the corresponding allowed mass term.	May be matrix-valued when several generations of F are present.
$m_{S_1}^2$	scalar mass-squared	$-m_{S_1}^2 S_1^\dagger S_1$	Quadratic scalar-potential parameter.
$m_{S_2}^2$	scalar mass-squared	$-m_{S_2}^2 S_2^\dagger S_2$	Quadratic scalar-potential parameter.
λ_H	SM Higgs quartic	$-\lambda_H (H^\dagger H)^2$	Overall sign follows the project's Lagrangian/potential convention.
λ_{S_1}	S_1 self quartic	$-\lambda_{S_1} (S_1^\dagger S_1)^2$	Singlet contraction of the scalar bilinears.
λ_{S_2}	S_2 self quartic	$-\lambda_{S_2} (S_2^\dagger S_2)^2$	Singlet contraction of the scalar bilinears.
λ_{H1}	Higgs- S_1 quartic	$-\lambda_{H1} (H^\dagger H)(S_1^\dagger S_1)$	Singlet-singlet mixed quartic.
λ_{H2}	Higgs- S_2 quartic	$-\lambda_{H2} (H^\dagger H)(S_2^\dagger S_2)$	Singlet-singlet mixed quartic.
λ_{12}	S_1 - S_2 quartic	$-\lambda_{12} (S_1^\dagger S_1)(S_2^\dagger S_2)$	Singlet-singlet mixed quartic.
λ_{T3}	T3 loop-closing quartic	$-\lambda_{T3} H H S_1 S_2 + \text{h.c.}$	Schematic field content. Conjugations and the invariant tensor are fixed by hypercharge and the $SU(2)$ representations; this is the quartic joining the two scalar lines to the two Higgs legs in T3.
$\lambda_{H1}^{\text{Adj}}$	adjoint Higgs- S_1 quartic	$-\lambda_{H1}^{\text{Adj}} (H^\dagger T^a H)(S_1^\dagger T^a S_1)$	Exists only when the relevant representations admit the adjoint bilinear.
$\lambda_{H2}^{\text{Adj}}$	adjoint Higgs- S_2 quartic	$-\lambda_{H2}^{\text{Adj}} (H^\dagger T^a H)(S_2^\dagger T^a S_2)$	Independent from the singlet contraction λ_{H2} .
$\lambda_{12}^{\text{Adj}}$	adjoint S_1 - S_2 quartic	$-\lambda_{12}^{\text{Adj}} (S_1^\dagger T^a S_1)(S_2^\dagger T^a S_2)$	Representation-dependent independent invariant.
$\lambda_{S_1}^{\text{Adj}}$	adjoint-type S_1 self quartic	$-\lambda_{S_1}^{\text{Adj}} (S_1^\dagger T^a S_1)(S_1^\dagger T^a S_1)$	Only appears where that self-contraction is independent/non-vanishing.
$\lambda_{S_2}^{\text{Adj}}$	adjoint-type S_2 self quartic	$-\lambda_{S_2}^{\text{Adj}} (S_2^\dagger T^a S_2)(S_2^\dagger T^a S_2)$	Only appears where that self-contraction is independent/non-vanishing.
$\lambda_{12}^{\text{Cross}}$	additional crossed S_1 - S_2 invariant	Model-dependent four-scalar contraction of $S_1, S_1^\dagger, S_2, S_2^\dagger$.	The exact Clebsch-Gordan/invariant-tensor contraction is representation dependent, so the report does not guess a universal component formula.
$\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	special quartic	$-\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} H^\dagger H^\dagger S_1^\dagger S_1^\dagger + \text{h.c.}$	Only registered when gauge invariance allows this field content.
$\lambda_{H^\dagger H^\dagger S_2 S_2}$	special quartic	$-\lambda_{H^\dagger H^\dagger S_2 S_2} H^\dagger H^\dagger S_2 S_2 + \text{h.c.}$	Representation/hypercharge dependent.
$\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$	special S_1 - S_2 quartic	$-\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} (S_1^\dagger)^2 (S_2^\dagger)^2 + \text{h.c.}$	Exact invariant contraction is model dependent.
$\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}$	special S_1 - S_2 quartic	$-\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}} S_1^\dagger S_2 (S_2^\dagger)^2 + \text{h.c.}$	Field-content notation mirrors the RGBeta coupling name.

Symbol	Meaning	Schematic Lagrangian term	Notes
$\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}$	special S_1 – S_2 quartic	$-\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} S_1 (S_1^{\dagger})^2 S_2^{\dagger} + \text{h.c.}$	Field-content notation mirrors the RGBeta coupling name.
$\lambda_{HH^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text{Cross}}$	special crossed Higgs–scalar A quartic	A crossed invariant with field content $H, H^{\dagger}, S_1^{\dagger}, S_2^{\dagger}$.	Exact $SU(2)$ tensor contraction depends on the representation assignment.

$$\beta_{g_Y}^{(1)}$$

$$16\pi^2\,\mu\frac{dg_Y}{d\mu}=\beta_{g_Y}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{g_Y}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{43g_Y^3}{6}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{43g_Y^3}{6}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{59g_Y^3}{6}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{59g_Y^3}{6}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{59g_Y^3}{6}$

$$\beta_{g_2}^{(1)}$$

$$16\pi^2\,\mu\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{g_2}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{17g_2^3}{6}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{7g_2^3}{6}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{g_2^3}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{g_2^3}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{g_2^3}{6}$

$$\beta_{g_3}^{(1)}$$

$$16\pi^2\,\mu\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{g_3}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-7g_3^3$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-7g_3^3$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-7g_3^3$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-7g_3^3$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-7g_3^3$

$$\beta_{Y_u}^{(1)}$$

$$16\pi^2 \mu \frac{dY_u}{d\mu} = \beta_{Y_u}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{Y_u}^{(1)}$	
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3}{2}$	$ \begin{aligned} & 3 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) (Y_u)_{ij} \\ & + \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) (Y_u)_{ij} \\ & + 3 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) (Y_u)_{ij} \\ & - \frac{3}{2} \left(\left(Y_d \left(Y_d \right)^\dagger Y_u \right)_{ij} - \left(Y_u \left(Y_u \right)^\dagger Y_u \right)_{ij} \right) \\ & - \frac{9}{4} g_2^2 (Y_u)_{ij} \\ & - 8 g_3^2 (Y_u)_{ij} \\ & - \frac{17}{12} g_Y^2 (Y_u)_{ij} \end{aligned} $
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{3}{2}$	$ \begin{aligned} & 3 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) (Y_u)_{ij} \\ & + \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) (Y_u)_{ij} \\ & + 3 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) (Y_u)_{ij} \\ & - \frac{3}{2} \left(\left(Y_d \left(Y_d \right)^\dagger Y_u \right)_{ij} - \left(Y_u \left(Y_u \right)^\dagger Y_u \right)_{ij} \right) \\ & - \frac{9}{4} g_2^2 (Y_u)_{ij} \\ & - 8 g_3^2 (Y_u)_{ij} \\ & - \frac{17}{12} g_Y^2 (Y_u)_{ij} \end{aligned} $
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}$	$ \begin{aligned} & 3 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) (Y_u)_{ij} \\ & + \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) (Y_u)_{ij} \\ & + 3 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) (Y_u)_{ij} \\ & - \frac{3}{2} \left(\left(Y_d \left(Y_d \right)^\dagger Y_u \right)_{ij} - \left(Y_u \left(Y_u \right)^\dagger Y_u \right)_{ij} \right) \\ & - \frac{9}{4} g_2^2 (Y_u)_{ij} \\ & - 8 g_3^2 (Y_u)_{ij} \\ & - \frac{17}{12} g_Y^2 (Y_u)_{ij} \end{aligned} $
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3}{2}$	$ \begin{aligned} & 3 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) (Y_u)_{ij} \\ & + \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) (Y_u)_{ij} \\ & + 3 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) (Y_u)_{ij} \\ & - \frac{3}{2} \left(\left(Y_d \left(Y_d \right)^\dagger Y_u \right)_{ij} - \left(Y_u \left(Y_u \right)^\dagger Y_u \right)_{ij} \right) \\ & - \frac{9}{4} g_2^2 (Y_u)_{ij} \\ & - 8 g_3^2 (Y_u)_{ij} \\ & - \frac{17}{12} g_Y^2 (Y_u)_{ij} \end{aligned} $

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{Y_u}^{(1)}$
								$3 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) \left(Y_u \right)_{ij}$ $+ \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) \left(Y_u \right)_{ij}$ $+ 3 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) \left(Y_u \right)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2} \left(\left(Y_d \left(Y_d \right)^\dagger Y_u \right)_{ij} - \left(Y_u \left(Y_u \right)^\dagger Y_u \right)_{ij} \right)$ $-\frac{9}{4} g_2^2 \left(Y_u \right)_{ij}$ $-8 g_3^2 \left(Y_u \right)_{ij}$ $-\frac{17}{12} g_Y^2 \left(Y_u \right)_{ij}$

$$16\pi^2 \mu \frac{dY_d}{d\mu} = \beta_{Y_d}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta Y_d^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$ \begin{aligned} & \text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)\left(Y_d\right)_{ij} \\ & +3 \text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\left(Y_d\right)_{ij} \\ & +3 \text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)\left(Y_d\right)_{ij} \\ & +\frac{3}{2}\left(\left(Y_d\left(Y_d\right)^{\dagger} Y_d\right)_{ij}-\left(Y_u\left(Y_u\right)^{\dagger} Y_d\right)_{ij}\right) \\ & -\frac{9}{4} g_2^2\left(Y_d\right)_{ij} \\ & -8 g_3^2\left(Y_d\right)_{ij} \\ & -\frac{5}{12} g_Y^2\left(Y_d\right)_{ij} \end{aligned} $
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$ \begin{aligned} & \text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)\left(Y_d\right)_{ij} \\ & +3 \text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\left(Y_d\right)_{ij} \\ & +3 \text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)\left(Y_d\right)_{ij} \\ & +\frac{3}{2}\left(\left(Y_d\left(Y_d\right)^{\dagger} Y_d\right)_{ij}-\left(Y_u\left(Y_u\right)^{\dagger} Y_d\right)_{ij}\right) \\ & -\frac{9}{4} g_2^2\left(Y_d\right)_{ij} \\ & -8 g_3^2\left(Y_d\right)_{ij} \\ & -\frac{5}{12} g_Y^2\left(Y_d\right)_{ij} \end{aligned} $
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$ \begin{aligned} & \text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)\left(Y_d\right)_{ij} \\ & +3 \text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\left(Y_d\right)_{ij} \\ & +3 \text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)\left(Y_d\right)_{ij} \\ & +\frac{3}{2}\left(\left(Y_d\left(Y_d\right)^{\dagger} Y_d\right)_{ij}-\left(Y_u\left(Y_u\right)^{\dagger} Y_d\right)_{ij}\right) \\ & -\frac{9}{4} g_2^2\left(Y_d\right)_{ij} \\ & -8 g_3^2\left(Y_d\right)_{ij} \\ & -\frac{5}{12} g_Y^2\left(Y_d\right)_{ij} \end{aligned} $
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$ \begin{aligned} & \text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)\left(Y_d\right)_{ij} \\ & +3 \text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\left(Y_d\right)_{ij} \\ & +3 \text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)\left(Y_d\right)_{ij} \\ & +\frac{3}{2}\left(\left(Y_d\left(Y_d\right)^{\dagger} Y_d\right)_{ij}-\left(Y_u\left(Y_u\right)^{\dagger} Y_d\right)_{ij}\right) \\ & -\frac{9}{4} g_2^2\left(Y_d\right)_{ij} \\ & -8 g_3^2\left(Y_d\right)_{ij} \\ & -\frac{5}{12} g_Y^2\left(Y_d\right)_{ij} \end{aligned} $

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta Y_d^{(1)}$
								$\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)\left(Y_d\right)_{ij}$ $+3\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\left(Y_d\right)_{ij}$ $+3\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)\left(Y_d\right)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}\left(\left(Y_d\left(Y_d\right)^{\dagger} Y_d\right)_{ij}-\left(Y_u\left(Y_u\right)^{\dagger} Y_d\right)_{ij}\right)$ $-\frac{9}{4} g_2^2\left(Y_d\right)_{ij}$ $-8 g_3^2\left(Y_d\right)_{ij}$ $-\frac{5}{12} g_Y^2\left(Y_d\right)_{ij}$

$$16\pi^2 \mu \frac{dY_e}{d\mu} = \beta_{Y_e}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{Y_e}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$ \begin{aligned} & 3 \text{Tr} \left(Y_d (Y_d)^\dagger \right) (Y_e)_{ij} \\ & + 3 \text{Tr} \left(Y_u (Y_u)^\dagger \right) (Y_e)_{ij} \\ & + \text{Tr} \left(Y_e (Y_e)^\dagger \right) (Y_e)_{ij} \\ & + \frac{1}{2} \left((y_1 (y_1)^\dagger Y_e)_{ij} + (y_2 (y_2)^\dagger Y_e)_{ij} + 3 \left(Y_e (Y_e)^\dagger Y_e \right)_{ij} \right) \\ & - \frac{9}{4} g_2^2 (Y_e)_{ij} \\ & - \frac{15}{4} g_Y^2 (Y_e)_{ij} \end{aligned} $
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$ \begin{aligned} & 3 \text{Tr} \left(Y_d (Y_d)^\dagger \right) (Y_e)_{ij} \\ & + 3 \text{Tr} \left(Y_u (Y_u)^\dagger \right) (Y_e)_{ij} \\ & + \text{Tr} \left(Y_e (Y_e)^\dagger \right) (Y_e)_{ij} \\ & + \frac{3}{4} \left((y_1 (y_1)^\dagger Y_e)_{ij} + (y_2 (y_2)^\dagger Y_e)_{ij} + 2 \left(Y_e (Y_e)^\dagger Y_e \right)_{ij} \right) \\ & - \frac{9}{4} g_2^2 (Y_e)_{ij} \\ & - \frac{15}{4} g_Y^2 (Y_e)_{ij} \end{aligned} $
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$ \begin{aligned} & 3 \text{Tr} \left(Y_d (Y_d)^\dagger \right) (Y_e)_{ij} \\ & + 3 \text{Tr} \left(Y_u (Y_u)^\dagger \right) (Y_e)_{ij} \\ & + \text{Tr} \left(Y_e (Y_e)^\dagger \right) (Y_e)_{ij} \\ & + \frac{1}{4} \left(2 \left(y_1 (y_1)^\dagger Y_e \right)_{ij} + 3 \left(y_2 (y_2)^\dagger Y_e \right)_{ij} + 6 \left(Y_e (Y_e)^\dagger Y_e \right)_{ij} \right) \\ & - \frac{9}{4} g_2^2 (Y_e)_{ij} \\ & - \frac{15}{4} g_Y^2 (Y_e)_{ij} \end{aligned} $
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$ \begin{aligned} & 3 \text{Tr} \left(Y_d (Y_d)^\dagger \right) (Y_e)_{ij} \\ & + 3 \text{Tr} \left(Y_u (Y_u)^\dagger \right) (Y_e)_{ij} \\ & + \text{Tr} \left(Y_e (Y_e)^\dagger \right) (Y_e)_{ij} \\ & + \frac{1}{4} \left(3 \left(y_1 (y_1)^\dagger Y_e \right)_{ij} + 2 \left(y_2 (y_2)^\dagger Y_e \right)_{ij} + 6 \left(Y_e (Y_e)^\dagger Y_e \right)_{ij} \right) \\ & - \frac{9}{4} g_2^2 (Y_e)_{ij} \\ & - \frac{15}{4} g_Y^2 (Y_e)_{ij} \end{aligned} $

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{Y_e}^{(1)}$
								$3 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) (Y_e)_{ij}$ $+ 3 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) (Y_e)_{ij}$ $+ \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) (Y_e)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \frac{3}{4} \left(\left(y_1 \left(y_1 \right)^\dagger Y_e \right)_{ij} + \left(y_2 \left(y_2 \right)^\dagger Y_e \right)_{ij} + 2 \left(Y_e \left(Y_e \right)^\dagger Y_e \right)_{ij} \right)$ $- \frac{9}{4} g_2^2 (Y_e)_{ij}$ $- \frac{15}{4} g_Y^2 (Y_e)_{ij}$

$\beta_{y_1}^{(1)}$

$$16\pi^2\mu\frac{dy_1}{d\mu}=\beta_{y_1}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{y_1}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{1}{2}(2\left(y_1\left(y_2\right)^{\dagger}y_2\right)_{ij}+\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ $-\frac{9}{4}g_2^2\left(y_1\right)_{ij}$ $-\frac{3}{4}g_Y^2\left(y_1\right)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{1}{4}(2\left(y_1\left(y_2\right)^{\dagger}y_2\right)_{ij}+3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+5\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ $-\frac{33}{4}g_2^2\left(y_1\right)_{ij}$ $-\frac{3}{4}g_Y^2\left(y_1\right)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$2\mathrm{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$ $+\frac{3}{4}\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}$ $+\frac{1}{2}\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}$ $+\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij}$ $-\frac{9}{2}g_2^2\left(y_1\right)_{ij}$ $-\frac{3}{2}g_Y^2\left(y_1\right)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\mathrm{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$ $+\frac{1}{2}((y_2\left(y_2\right)^{\dagger}y_1)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ $-\frac{9}{2}g_2^2\left(y_1\right)_{ij}$ $-\frac{3}{2}g_Y^2\left(y_1\right)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\mathrm{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$ $+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+6\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ $-\frac{9}{2}g_2^2\left(y_1\right)_{ij}$ $-\frac{3}{2}g_Y^2\left(y_1\right)_{ij}$

$$\beta_{y_2}^{(1)}$$

$$16\pi^2\mu\frac{dy_2}{d\mu}=\beta_{y_2}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{y_2}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\begin{aligned} & \text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij} \\ & +\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij} \\ & +\frac{1}{2}\left(\left(y_1\left(y_1\right)^{\dagger} y_2\right)_{ij}+2\left(y_2\left(y_1\right)^{\dagger} y_1\right)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger} y_2\right)_{ij}+3\left(y_2\left(y_2\right)^{\dagger} y_2\right)_{ij}\right) \\ & -\frac{9}{4} g_2^2\left(y_2\right)_{ij} \\ & -\frac{3}{4} g_Y^2\left(y_2\right)_{ij} \end{aligned}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\begin{aligned} & \frac{3}{2} \text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij} \\ & +\frac{3}{2} \text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij} \\ & +\frac{1}{4}\left(3\left(y_1\left(y_1\right)^{\dagger} y_2\right)_{ij}+2\left(y_2\left(y_1\right)^{\dagger} y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger} y_2\right)_{ij}+5\left(y_2\left(y_2\right)^{\dagger} y_2\right)_{ij}\right) \\ & -\frac{33}{4} g_2^2\left(y_2\right)_{ij} \\ & -\frac{3}{4} g_Y^2\left(y_2\right)_{ij} \end{aligned}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\begin{aligned} & \text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij} \\ & +\frac{1}{2}\left(\left(y_1\left(y_1\right)^{\dagger} y_2\right)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger} y_2\right)_{ij}+3\left(y_2\left(y_2\right)^{\dagger} y_2\right)_{ij}\right) \\ & -\frac{9}{2} g_2^2\left(y_2\right)_{ij} \\ & -\frac{3}{2} g_Y^2\left(y_2\right)_{ij} \end{aligned}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\begin{aligned} & 2 \text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij} \\ & +\frac{3}{4}\left(y_1\left(y_1\right)^{\dagger} y_2\right)_{ij} \\ & +\frac{1}{2}\left(Y_e\left(Y_e\right)^{\dagger} y_2\right)_{ij} \\ & +\left(y_2\left(y_2\right)^{\dagger} y_2\right)_{ij} \\ & -\frac{9}{2} g_2^2\left(y_2\right)_{ij} \\ & -\frac{3}{2} g_Y^2\left(y_2\right)_{ij} \end{aligned}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\begin{aligned} & \text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij} \\ & +\frac{1}{4}\left(3\left(y_1\left(y_1\right)^{\dagger} y_2\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger} y_2\right)_{ij}+6\left(y_2\left(y_2\right)^{\dagger} y_2\right)_{ij}\right) \\ & -\frac{9}{2} g_2^2\left(y_2\right)_{ij} \\ & -\frac{3}{2} g_Y^2\left(y_2\right)_{ij} \end{aligned}$

$$\beta_{M_F}^{(1)}$$

$$16\pi^2\mu\frac{dM_F}{d\mu}=\beta_{M_F}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{M_F}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\left((M_F)^T(y_1)^\dagger y_1\right)_{ij}$
								$+\left((y_1)^T(y_1)^*M_F\right)_{ij}$
								$+\left((M_F)^T(y_2)^\dagger y_2\right)_{ij}$
								$+\left((y_2)^T(y_2)^*M_F\right)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{1}{2}\left(\left((M_F)^T(y_1)^\dagger y_1\right)_{ij}+\left((y_1)^T(y_1)^*M_F\right)_{ij}+\left((M_F)^T(y_2)^\dagger y_2\right)_{ij}+\left((y_2)^T(y_2)^*M_F\right)_{ij}\right)$
								$-6g_2^2\left(\left((M_F)^T\right)_{ij}+(M_F)_{ij}\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{1}{4}\left(2\left((y_1)^T(y_1)^*M_F\right)_{ij}+3\left(M_F(y_2)^\dagger y_2\right)_{ij}\right)$
								$-\frac{9}{2}g_2^2(M_F)_{ij}$
								$-\frac{3}{2}g_Y^2(M_F)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{1}{4}\left(3\left((y_1)^T(y_1)^*M_F\right)_{ij}+2\left(M_F(y_2)^\dagger y_2\right)_{ij}\right)$
								$-\frac{9}{2}g_2^2(M_F)_{ij}$
								$-\frac{3}{2}g_Y^2(M_F)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{3}{4}\left(\left((y_1)^T(y_1)^*M_F\right)_{ij}+\left(M_F(y_2)^\dagger y_2\right)_{ij}\right)$
								$-\frac{9}{2}g_2^2(M_F)_{ij}$
								$-\frac{3}{2}g_Y^2(M_F)_{ij}$

$$\beta_{m_{S_1}^2}^{(1)}$$

$$16\pi^2 \mu \frac{dm_{S_1}^2}{d\mu} = \beta_{m_{S_1}^2}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta \frac{(1)}{m_{S_1}^2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$m_{S_1}^2 (2 \text{Tr} (y_1 (y_1)^\dagger) + 6\lambda_{S_1})$ $-8 \text{Tr} (M_F (M_F)^* (y_1)^T (y_1)^*)$ $-8 \text{Tr} (M_F (y_1)^\dagger y_1 (M_F)^*)$ $-\frac{9g_2^2 m_{S_1}^2}{2}$ $-\frac{3g_Y^2 m_{S_1}^2}{2}$ $+4\lambda_{12} m_{S_2}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$m_{S_1}^2 (3 \text{Tr} (y_1 (y_1)^\dagger) + 6\lambda_{S_1})$ $-12 \text{Tr} (M_F (M_F)^* (y_1)^T (y_1)^*)$ $-12 \text{Tr} (M_F (y_1)^\dagger y_1 (M_F)^*)$ $-\frac{9g_2^2 m_{S_1}^2}{2}$ $-\frac{3g_Y^2 m_{S_1}^2}{2}$ $+4\lambda_{12} m_{S_2}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$m_{S_1}^2 (4 \text{Tr} (y_1 (y_1)^\dagger) + 4\lambda_{S_1})$ $-8 \text{Tr} (M_F (M_F)^\dagger (y_1)^T (y_1)^*)$ $+6\lambda_{12} m_{S_2}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$m_{S_1}^2 (2 \text{Tr} (y_1 (y_1)^\dagger) + 8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$ $-4 \text{Tr} (M_F (M_F)^\dagger (y_1)^T (y_1)^*)$ $-12g_2^2 m_{S_1}^2$ $-6g_Y^2 m_{S_1}^2$ $+2\lambda_{12} m_{S_2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$m_{S_1}^2 (2 \text{Tr} (y_1 (y_1)^\dagger) + 8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$ $-4 \text{Tr} (M_F (M_F)^\dagger (y_1)^T (y_1)^*)$ $-12g_2^2 m_{S_1}^2$ $+m_{S_2}^2 (6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$

$$\beta_{m_{S_2}^2}^{(1)}$$

$$16\pi^2 \mu \frac{dm_{S_2}^2}{d\mu} = \beta_{m_{S_2}^2}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta m_{S_2}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$m_{S_2}^2 (2 \text{Tr} (y_2 (y_2)^\dagger) + 6\lambda_{S_2})$ $-8 \text{Tr} (M_F (M_F)^* (y_2)^T (y_2)^*)$ $-8 \text{Tr} (M_F (y_2)^\dagger y_2 (M_F)^*)$ $-\frac{9g_2^2 m_{S_2}^2}{2}$ $-\frac{3g_Y^2 m_{S_2}^2}{2}$ $+4\lambda_{12} m_{S_1}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$m_{S_2}^2 (3 \text{Tr} (y_2 (y_2)^\dagger) + 6\lambda_{S_2})$ $-12 \text{Tr} (M_F (M_F)^* (y_2)^T (y_2)^*)$ $-12 \text{Tr} (M_F (y_2)^\dagger y_2 (M_F)^*)$ $-\frac{9g_2^2 m_{S_2}^2}{2}$ $-\frac{3g_Y^2 m_{S_2}^2}{2}$ $+4\lambda_{12} m_{S_1}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$m_{S_2}^2 (2 \text{Tr} (y_2 (y_2)^\dagger) + 8\lambda_{S_2} + 4\lambda_{S_2}^{\text{Adj}})$ $-4 \text{Tr} (M_F (y_2)^\dagger y_2 (M_F)^\dagger)$ $-12g_2^2 m_{S_2}^2$ $-6g_Y^2 m_{S_2}^2$ $+2\lambda_{12} m_{S_1}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$m_{S_2}^2 (4 \text{Tr} (y_2 (y_2)^\dagger) + 4\lambda_{S_2})$ $-8 \text{Tr} (M_F (y_2)^\dagger y_2 (M_F)^\dagger)$ $+6\lambda_{12} m_{S_1}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$m_{S_2}^2 (2 \text{Tr} (y_2 (y_2)^\dagger) + 8\lambda_{S_2} + 4\lambda_{S_2}^{\text{Adj}})$ $-4 \text{Tr} (M_F (y_2)^\dagger y_2 (M_F)^\dagger)$ $-12g_2^2 m_{S_2}^2$ $-6g_Y^2 m_{S_2}^2$ $+m_{S_1}^2 (6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$

$$\beta_{\lambda_H}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_H}{d\mu}=\beta_{\lambda_H}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_H}^{(1)}$
								$\lambda_H(12\text{Tr}\Big(Y_d\;(Y_d)^\dagger\Big)+4\text{Tr}\Big(Y_e\;(Y_e)^\dagger\Big)+12\text{Tr}\Big(Y_u\;(Y_u)^\dagger\Big))$ $-12\text{Tr}\Big(Y_d\;(Y_d)^\dagger\;Y_d\;(Y_d)^\dagger\Big)$ $-4\text{Tr}\Big(Y_e\;(Y_e)^\dagger\;Y_e\;(Y_e)^\dagger\Big)$ $-12\text{Tr}\Big(Y_u\;(Y_u)^\dagger\;Y_u\;(Y_u)^\dagger\Big)$ $+8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ $+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$ $+8\lambda_{H^\dagger H^\dagger S_2 S_2}\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$ $+4\lambda_{T3}(\lambda_{T3})^*$ $+\frac{9g_2^4}{4}$ $+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$ $-9g_2^2\lambda_H$ $+\frac{3g_Y^4}{4}$ $+12\lambda_H^2$ $+4\lambda_{H1}^2$ $+\frac{\lambda_{H1}^{\text{Adj}2}}{4}$ $+4\lambda_{H2}^2$ $+\frac{\lambda_{H2}^{\text{Adj}2}}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_H}^{(1)}$
								$ \begin{aligned} &\lambda_H(12\text{Tr}\left(Y_d(Y_d)^\dagger\right)+4\text{Tr}\left(Y_e(Y_e)^\dagger\right)+12\text{Tr}\left(Y_u(Y_u)^\dagger\right)) \\ &\quad -12\text{Tr}\left(Y_d(Y_d)^\dagger Y_d(Y_d)^\dagger\right) \\ &\quad -4\text{Tr}\left(Y_e(Y_e)^\dagger Y_e(Y_e)^\dagger\right) \\ &\quad -12\text{Tr}\left(Y_u(Y_u)^\dagger Y_u(Y_u)^\dagger\right) \\ &\quad +8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^* \\ &\quad +4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* \\ &\quad +8\lambda_{H^\dagger H^\dagger S_2 S_2}\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^* \\ &\quad +4\lambda_{T3}(\lambda_{T3})^* \\ &\quad +\frac{9g_2^4}{4} \\ &\quad +g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right) \\ &\quad -9g_2^2\lambda_H \\ &\quad +\frac{3g_Y^4}{4} \\ &\quad +12\lambda_H^2 \\ &\quad +4\lambda_{H1}^2 \\ &\quad +\frac{\lambda_{H1}^{\text{Adj}2}}{4} \\ &\quad +4\lambda_{H2}^2 \\ &\quad +\frac{\lambda_{H2}^{\text{Adj}2}}{4} \end{aligned} $
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	
								$ \begin{aligned} &\lambda_H(12\text{Tr}\left(Y_d(Y_d)^\dagger\right)+4\text{Tr}\left(Y_e(Y_e)^\dagger\right)+12\text{Tr}\left(Y_u(Y_u)^\dagger\right)) \\ &\quad -12\text{Tr}\left(Y_d(Y_d)^\dagger Y_d(Y_d)^\dagger\right) \\ &\quad -4\text{Tr}\left(Y_e(Y_e)^\dagger Y_e(Y_e)^\dagger\right) \\ &\quad -12\text{Tr}\left(Y_u(Y_u)^\dagger Y_u(Y_u)^\dagger\right) \\ &\quad +4\lambda_{T3}(\lambda_{T3})^* \\ &\quad +\frac{9g_2^4}{4} \\ &\quad +g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right) \\ &\quad -9g_2^2\lambda_H \\ &\quad +\frac{3g_Y^4}{4} \\ &\quad +12\lambda_H^2 \\ &\quad +2\lambda_{H1}^2 \\ &\quad +6\lambda_{H2}^2 \\ &\quad +3\lambda_{H2}^{\text{Adj}2} \end{aligned} $
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta^{(1)}_{\lambda_H}$
								$\lambda_H(12 \text{Tr}\left(Y_d (Y_d)^\dagger\right) + 4 \text{Tr}\left(Y_e (Y_e)^\dagger\right) + 12 \text{Tr}\left(Y_u (Y_u)^\dagger\right))$ $-12 \text{Tr}\left(Y_d (Y_d)^\dagger Y_d (Y_d)^\dagger\right)$ $-4 \text{Tr}\left(Y_e (Y_e)^\dagger Y_e (Y_e)^\dagger\right)$ $-12 \text{Tr}\left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger\right)$ $+4\lambda_{T3} (\lambda_{T3})^*$ $+\frac{9g_2^4}{4}$ $+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H\right)$ $-9g_2^2 \lambda_H$ $+\frac{3g_Y^4}{4}$ $+12\lambda_H^2$ $+6\lambda_{H1}^2$ $+3\lambda_{H1}^{\text{Adj}2}$ $+2\lambda_{H2}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_H(12 \text{Tr}\left(Y_d (Y_d)^\dagger\right) + 4 \text{Tr}\left(Y_e (Y_e)^\dagger\right) + 12 \text{Tr}\left(Y_u (Y_u)^\dagger\right))$ $-12 \text{Tr}\left(Y_d (Y_d)^\dagger Y_d (Y_d)^\dagger\right)$ $-4 \text{Tr}\left(Y_e (Y_e)^\dagger Y_e (Y_e)^\dagger\right)$ $-12 \text{Tr}\left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger\right)$ $+4\lambda_{T3} (\lambda_{T3})^*$ $+\frac{9g_2^4}{4}$ $+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H\right)$ $-9g_2^2 \lambda_H$ $+\frac{3g_Y^4}{4}$ $+12\lambda_H^2$ $+6\lambda_{H1}^2$ $+3\lambda_{H1}^{\text{Adj}2}$ $+2\lambda_{H2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_H(12 \text{Tr}\left(Y_d (Y_d)^\dagger\right) + 4 \text{Tr}\left(Y_e (Y_e)^\dagger\right) + 12 \text{Tr}\left(Y_u (Y_u)^\dagger\right))$ $-12 \text{Tr}\left(Y_d (Y_d)^\dagger Y_d (Y_d)^\dagger\right)$ $-4 \text{Tr}\left(Y_e (Y_e)^\dagger Y_e (Y_e)^\dagger\right)$ $-12 \text{Tr}\left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger\right)$ $+4\lambda_{T3} (\lambda_{T3})^*$ $+\frac{9g_2^4}{4}$ $+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H\right)$ $-9g_2^2 \lambda_H$ $+\frac{3g_Y^4}{4}$ $+12\lambda_H^2$ $+6\lambda_{H1}^2$ $+3\lambda_{H1}^{\text{Adj}2}$ $+6\lambda_{H2}^2$ $+3\lambda_{H2}^{\text{Adj}2}$

$$\beta_{\lambda_{S_1}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{d\lambda_{S_1}}{d\mu}=\beta_{\lambda_{S_1}}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{S_1}}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$ \begin{aligned} &4\lambda_{S_1}\,\mathrm{Tr}\Big(y_1\,(y_1)^\dagger\Big)\\ &+\lambda_{S_1S_1^\dagger{}^2S_2^\dagger}\big(2\,\mathrm{Tr}\Big(y_1\,(y_2)^\dagger\Big)+24\,\Big(\lambda_{S_1S_1^\dagger{}^2S_2^\dagger}\Big)^*\big)\\ &+2\,\Big(\lambda_{S_1S_1^\dagger{}^2S_2^\dagger}\Big)^*\mathrm{Tr}\Big(y_2\,(y_1)^\dagger\Big)\\ &-4\,\mathrm{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)\\ &+8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\Big(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\Big)^*\\ &+8\lambda_{S_1^\dagger{}^2S_2^\dagger{}^2}\Big(\lambda_{S_1^\dagger{}^2S_2^\dagger{}^2}\Big)^*\\ &\quad+\frac{9g_2^4}{4}\\ &+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_{S_1}\right)\\ &\quad-9g_2^2\lambda_{S_1}\\ &\quad+\frac{3g_Y^4}{4}\\ &\quad+4\lambda_{12}^2\\ &\quad+\frac{\lambda_{12}^{\mathrm{Adj}^2}}{4}\\ &\quad+4\lambda_{H1}^2\\ &\quad+\frac{\lambda_{H1}^{\mathrm{Adj}^2}}{4}\\ &\quad+12\lambda_{S_1}^2 \end{aligned} $

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{S_1}}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$ \begin{aligned} & 6\lambda_{S_1} \text{Tr}\left(y_1 (y_1)^\dagger\right) \\ & + \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \left(3 \text{Tr}\left(y_1 (y_2)^\dagger\right) + 24 \left(\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \right)^* \right) \\ & + 3 \left(\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \right)^* \text{Tr}\left(y_2 (y_1)^\dagger\right) \\ & - 5 \text{Tr}\left(y_1 (y_1)^\dagger y_1 (y_1)^\dagger\right) \\ & + 8\lambda_{H^\dagger H^\dagger S_1^{\dagger} S_1^{\dagger}} \left(\lambda_{H^\dagger H^\dagger S_1^{\dagger} S_1^{\dagger}} \right)^* \\ & + 8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^* \\ & + \frac{9g_2^4}{4} \\ & + g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_1} \right) \\ & - 9g_2^2 \lambda_{S_1} \\ & + \frac{3g_Y^4}{4} \\ & + 4\lambda_{12}^2 \\ & + \frac{\lambda_{12}^{\text{Adj}2}}{4} \\ & + 4\lambda_{H1}^2 \\ & + \frac{\lambda_{H1}^{\text{Adj}2}}{4} \\ & + 12\lambda_{S_1}^2 \end{aligned} $
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$ \begin{aligned} & 8\lambda_{S_1} \text{Tr}\left(y_1 (y_1)^\dagger\right) \\ & - 8 \text{Tr}\left(y_1 (y_1)^\dagger y_1 (y_1)^\dagger\right) \\ & + 6\lambda_{12}^2 \\ & + 4\lambda_{H1}^2 \\ & + 10\lambda_{S_1}^2 \end{aligned} $
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$ \begin{aligned} & \lambda_{S_1} (4 \text{Tr}\left(y_1 (y_1)^\dagger\right) + 14\lambda_{S_1}^{\text{Adj}}) \\ & + 2\lambda_{S_1}^{\text{Adj}} \text{Tr}\left(y_1 (y_1)^\dagger\right) \\ & - 3 \text{Tr}\left(y_1 (y_1)^\dagger y_1 (y_1)^\dagger\right) \\ & + 21g_2^4 \\ & + g_Y^2 (12g_2^2 - 12\lambda_{S_1} - 6\lambda_{S_1}^{\text{Adj}}) \\ & + g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}}) \\ & + 12g_Y^4 \\ & + 2\lambda_{12}^2 \\ & + 4\lambda_{H1}^2 \\ & + \frac{3\lambda_{H1}^{\text{Adj}2}}{2} \\ & + 14\lambda_{S_1}^2 \\ & + 26\lambda_{S_1}^{\text{Adj}2} \end{aligned} $

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta \lambda_{S_1}^{(1)}$	
									$\lambda_{S_1} (4 \operatorname{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + 14 \lambda_{S_1}^{\text{Adj}})$ $+ 2 \lambda_{S_1}^{\text{Adj}} \operatorname{Tr} \left(y_1 \left(y_1 \right)^\dagger \right)$ $- 3 \operatorname{Tr} \left(y_1 \left(y_1 \right)^\dagger y_1 \left(y_1 \right)^\dagger \right)$ $+ 21 g_2^4$ $+ g_2^2 (-24 \lambda_{S_1} - 12 \lambda_{S_1}^{\text{Adj}})$ $+ 6 \lambda_{12}^2$ $+ 8 \lambda_{12} \lambda_{12}^{\text{Cross}}$ $+ 6 \lambda_{12}^{\text{Adj}^2}$ $+ 6 \lambda_{12}^{\text{Cross}^2}$ $+ 4 \lambda_{H1}^2$ $+ \frac{3 \lambda_{H1}^{\text{Adj}^2}}{2}$ $+ 14 \lambda_{S_1}^2$ $+ 26 \lambda_{S_1}^{\text{Adj}^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$		

$$\beta_{\lambda_{S_2}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{d\lambda_{S_2}}{d\mu}=\beta_{\lambda_{S_2}}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{S_2}}^{(1)}$
								$\lambda_{S_1^\dagger S_2 S_2^\dagger 2} (2\, \text{Tr}\Big(y_1\, (y_2)^\dagger\Big) + 24\, \Big(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\Big)^*)$ $+ 2\, \Big(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\Big)^* \, \text{Tr}\Big(y_2\, (y_1)^\dagger\Big)$ $+ 4\lambda_{S_2}\, \text{Tr}\Big(y_2\, (y_2)^\dagger\Big)$ $- 4\, \text{Tr}\Big(y_2\, (y_2)^\dagger\, y_2\, (y_2)^\dagger\Big)$ $+ 8\lambda_{H^\dagger H^\dagger S_2 S_2}\, \Big(\lambda_{H^\dagger H^\dagger S_2 S_2}\Big)^*$ $+ 8\lambda_{S_1^\dagger 2 S_2^\dagger 2}\, \Big(\lambda_{S_1^\dagger 2 S_2^\dagger 2}\Big)^*$ $+ \frac{9g_2^4}{4}$ $+ g_Y^2\, \Big(\frac{3g_2^2}{2} - 3\lambda_{S_2}\Big)$ $- 9g_2^2\lambda_{S_2}$ $+ \frac{3g_Y^4}{4}$ $+ 4\lambda_{12}^2$ $+ \frac{\lambda_{12}^{\text{Adj}2}}{4}$ $+ 4\lambda_{H2}^2$ $+ \frac{\lambda_{H2}^{\text{Adj}2}}{4}$ $+ 12\lambda_{S_2}^2$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta \lambda_{S_2}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$ \begin{aligned} & \lambda_{S_1^\dagger S_2 S_2^\dagger 2} (3 \text{Tr} (y_1 (y_2)^\dagger) + 24 \left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^*) \\ & + 3 \left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* \text{Tr} (y_2 (y_1)^\dagger) \\ & + 6 \lambda_{S_2} \text{Tr} (y_2 (y_2)^\dagger) \\ & - 5 \text{Tr} (y_2 (y_2)^\dagger y_2 (y_2)^\dagger) \\ & + 8 \lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* \\ & + 8 \lambda_{S_1^\dagger 2 S_2^\dagger 2} \left(\lambda_{S_1^\dagger 2 S_2^\dagger 2} \right)^* \\ & + \frac{9 g_2^4}{4} \\ & + g_Y^2 \left(\frac{3 g_2^2}{2} - 3 \lambda_{S_2} \right) \\ & - 9 g_2^2 \lambda_{S_2} \\ & + \frac{3 g_Y^4}{4} \\ & + 4 \lambda_{12}^2 \\ & + \frac{\lambda_{12}^{\text{Adj}2}}{4} \\ & + 4 \lambda_{H2}^2 \\ & + \frac{\lambda_{H2}^{\text{Adj}2}}{4} \\ & + 12 \lambda_{S_2}^2 \end{aligned} $
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$ \begin{aligned} & \lambda_{S_2} (4 \text{Tr} (y_2 (y_2)^\dagger) + 14 \lambda_{S_2}^{\text{Adj}}) \\ & + 2 \lambda_{S_2}^{\text{Adj}} \text{Tr} (y_2 (y_2)^\dagger) \\ & - 3 \text{Tr} (y_2 (y_2)^\dagger y_2 (y_2)^\dagger) \\ & + 21 g_2^4 \\ & + g_Y^2 (12 g_2^2 - 12 \lambda_{S_2} - 6 \lambda_{S_2}^{\text{Adj}}) \\ & + g_2^2 (-24 \lambda_{S_2} - 12 \lambda_{S_2}^{\text{Adj}}) \\ & + 12 g_Y^4 \\ & + 2 \lambda_{12}^2 \\ & + 4 \lambda_{H2}^2 \\ & + \frac{3 \lambda_{H2}^{\text{Adj}2}}{2} \\ & + 14 \lambda_{S_2}^2 \\ & + 26 \lambda_{S_2}^{\text{Adj}2} \end{aligned} $
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$ \begin{aligned} & 8 \lambda_{S_2} \text{Tr} (y_2 (y_2)^\dagger) \\ & - 8 \text{Tr} (y_2 (y_2)^\dagger y_2 (y_2)^\dagger) \\ & + 6 \lambda_{12}^2 \\ & + 4 \lambda_{H2}^2 \\ & + 10 \lambda_{S_2}^2 \end{aligned} $

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta^{(1)}_{\lambda_{S_2}}$	
									$\lambda_{S_2} (4 \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) + 14 \lambda_{S_2}^{\text{Adj}})$ $+ 2 \lambda_{S_2}^{\text{Adj}} \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right)$ $- 3 \text{Tr} \left(y_2 \left(y_2 \right)^\dagger y_2 \left(y_2 \right)^\dagger \right)$ $+ 21 g_2^4$ $+ g_Y^2 \left(12 g_2^2 - 12 \lambda_{S_2} - 6 \lambda_{S_2}^{\text{Adj}} \right)$ $+ g_2^2 \left(-24 \lambda_{S_2} - 12 \lambda_{S_2}^{\text{Adj}} \right)$ $+ 12 g_Y^4$ $+ 6 \lambda_{12}^2$ $+ 8 \lambda_{12} \lambda_{12}^{\text{Cross}}$ $+ 6 \lambda_{12}^{\text{Adj}2}$ $+ 6 \lambda_{12}^{\text{Cross}2}$ $+ 4 \lambda_{H2}^2$ $+ \frac{3 \lambda_{H2}^{\text{Adj}2}}{2}$ $+ 14 \lambda_{S_2}^2$ $+ 26 \lambda_{S_2}^{\text{Adj}2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$		

$$\beta_{\lambda_{H1}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{d\lambda_{H1}}{d\mu}=\beta_{\lambda_{H1}}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{H1}}^{(1)}$
								$\lambda_{H1}(2\,\text{Tr}\Big(y_1\,\,(y_1)^\dagger\Big)+6\,\text{Tr}\Big(Y_d\,\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,\,(Y_u)^\dagger\Big))$ $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\,\text{Tr}\Big(y_1\,\,(y_2)^\dagger\Big)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*\right)$ $+\frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\,\text{Tr}\Big(y_2\,\,(y_1)^\dagger\Big)$ $-2\,\text{Tr}\Big(y_1\,\,(y_1)^\dagger\,Y_e\,\,(Y_e)^\dagger\Big)$ $+24\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ $+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$ $+6\lambda_{T3}\,(\lambda_{T3})^*$ $+\frac{9g_2^4}{4}$ $-9g_2^2\lambda_{H1}$ $+\frac{3g_Y^4}{4}$ $-3g_Y^2\lambda_{H1}$ $+4\lambda_{12}\lambda_{H2}$ $+6\lambda_H\lambda_{H1}$ $+4\lambda_{H1}^2$ $+6\lambda_{H1}\lambda_{S_1}$ $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta \lambda_{H1}^{(1)}$
								$\lambda_{H1} (3 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger))$ $+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\frac{3}{4} \text{Tr} (y_1 (y_2)^\dagger) + 4 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + 3 \left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \right)^* \right)$ $+ \frac{3}{4} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr} (y_2 (y_1)^\dagger)$ $- 3 \text{Tr} (y_1 (y_1)^\dagger Y_e (Y_e)^\dagger)$ $+ 24 \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$ $+ 3 \lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ $+ 6 \lambda_{T3} (\lambda_{T3})^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \frac{9 g_2^4}{4}$ $- 9 g_2^2 \lambda_{H1}$ $+ \frac{3 g_Y^4}{4}$ $- 3 g_Y^2 \lambda_{H1}$ $+ 4 \lambda_{12} \lambda_{H2}$ $+ 6 \lambda_H \lambda_{H1}$ $+ 4 \lambda_{H1}^2$ $+ 6 \lambda_{H1} \lambda_{S_1}$ $+ \frac{3 \lambda_{H1}^{\text{Adj}^2}}{4}$
								$\lambda_{H1} (4 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger))$ $- 4 \text{Tr} (y_1 (y_1)^\dagger Y_e (Y_e)^\dagger)$ $+ 12 \lambda_{T3} (\lambda_{T3})^*$ $- \frac{9 g_2^2 \lambda_{H1}}{2}$ $- \frac{3 g_Y^2 \lambda_{H1}}{2}$ $+ 6 \lambda_{12} \lambda_{H2}$ $+ 6 \lambda_H \lambda_{H1}$ $+ 4 \lambda_{H1}^2$ $+ 4 \lambda_{H1} \lambda_{S_1}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{H1}}^{(1)}$
								$\lambda_{H1} (2 \text{Tr} \left(y_1 (y_1)^\dagger \right) + 6 \text{Tr} \left(Y_d (Y_d)^\dagger \right) + 2 \text{Tr} \left(Y_e (Y_e)^\dagger \right) + 6 \text{Tr} \left(Y_u (Y_u)^\dagger \right) + 4 \lambda_{S_1}^{\text{Adj}})$ $- 2 \text{Tr} \left(y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$ $+ 4 \lambda_{T3} (\lambda_{T3})^*$ $+ 6 g_2^4$ $- \frac{33 g_2^2 \lambda_{H1}}{2}$ $+ 3 g_Y^4$ $- \frac{15 g_Y^2 \lambda_{H1}}{2}$ $+ 2 \lambda_{12} \lambda_{H2}$ $+ 6 \lambda_H \lambda_{H1}$ $+ 4 \lambda_{H1}^2$ $+ 8 \lambda_{H1} \lambda_{S_1}$ $+ 6 \lambda_{H1}^{\text{Adj}2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{H1} (2 \text{Tr} \left(y_1 (y_1)^\dagger \right) + 6 \text{Tr} \left(Y_d (Y_d)^\dagger \right) + 2 \text{Tr} \left(Y_e (Y_e)^\dagger \right) + 6 \text{Tr} \left(Y_u (Y_u)^\dagger \right) + 4 \lambda_{S_1}^{\text{Adj}})$ $- 2 \text{Tr} \left(y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$ $+ 4 \lambda_{T3} (\lambda_{T3})^*$ $+ 6 g_2^4$ $- \frac{33 g_2^2 \lambda_{H1}}{2}$ $- \frac{3 g_Y^2 \lambda_{H1}}{2}$ $+ \lambda_{H2} (6 \lambda_{12} + 4 \lambda_{12}^{\text{Cross}})$ $+ 6 \lambda_H \lambda_{H1}$ $+ 4 \lambda_{H1}^2$ $+ 8 \lambda_{H1} \lambda_{S_1}$ $+ 6 \lambda_{H1}^{\text{Adj}2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{H1} (2 \text{Tr} \left(y_1 (y_1)^\dagger \right) + 6 \text{Tr} \left(Y_d (Y_d)^\dagger \right) + 2 \text{Tr} \left(Y_e (Y_e)^\dagger \right) + 6 \text{Tr} \left(Y_u (Y_u)^\dagger \right) + 4 \lambda_{S_1}^{\text{Adj}})$ $- 2 \text{Tr} \left(y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$ $+ 4 \lambda_{T3} (\lambda_{T3})^*$ $+ 6 g_2^4$ $- \frac{33 g_2^2 \lambda_{H1}}{2}$ $- \frac{3 g_Y^2 \lambda_{H1}}{2}$ $+ \lambda_{H2} (6 \lambda_{12} + 4 \lambda_{12}^{\text{Cross}})$ $+ 6 \lambda_H \lambda_{H1}$ $+ 4 \lambda_{H1}^2$ $+ 8 \lambda_{H1} \lambda_{S_1}$ $+ 6 \lambda_{H1}^{\text{Adj}2}$

$$\beta_{\lambda_{H2}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{d\lambda_{H2}}{d\mu}=\beta_{\lambda_{H2}}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{H2}}^{(1)}$
								$\lambda_{H2}(2\,\text{Tr}\Big(y_2\,(y_2)^\dagger\Big)+6\,\text{Tr}\Big(Y_d\,(Y_d)^\dagger\Big)+2\,\text{Tr}\Big(Y_e\,(Y_e)^\dagger\Big)+6\,\text{Tr}\Big(Y_u\,(Y_u)^\dagger\Big))$ $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\,\text{Tr}\Big(y_1\,(y_2)^\dagger\Big)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1^\dagger S_2 S_2^\dagger}^2\right)^*\right)$ $+\frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\,\text{Tr}\Big(y_2\,(y_1)^\dagger\Big)$ $-2\,\text{Tr}\Big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\Big)$ $+3\lambda_{S_1^\dagger S_2 S_2^\dagger}^2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$ $+24\lambda_{H^\dagger H^\dagger S_2 S_2}\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$ $+6\lambda_{T3}\,(\lambda_{T3})^*$ $+\frac{9g_2^4}{4}$ $-9g_2^2\lambda_{H2}$ $+\frac{3g_Y^4}{4}$ $-3g_Y^2\lambda_{H2}$ $+4\lambda_{12}\lambda_{H1}$ $+6\lambda_H\lambda_{H2}$ $+4\lambda_{H2}^2$ $+6\lambda_{H2}\lambda_{S_2}$ $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta \lambda_{H2}^{(1)}$
								$\lambda_{H2} (3 \text{Tr} \left(y_2 (y_2)^\dagger \right) + 6 \text{Tr} \left(Y_d (Y_d)^\dagger \right) + 2 \text{Tr} \left(Y_e (Y_e)^\dagger \right) + 6 \text{Tr} \left(Y_u (Y_u)^\dagger \right))$ $+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\frac{3}{4} \text{Tr} \left(y_1 (y_2)^\dagger \right) + 4 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + 3 \left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* \right)$ $+ \frac{3}{4} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr} \left(y_2 (y_1)^\dagger \right)$ $- 3 \text{Tr} \left(y_2 (y_2)^\dagger Y_e (Y_e)^\dagger \right)$ $+ 3 \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ $+ 24 \lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$ $+ 6 \lambda_{T3} (\lambda_{T3})^*$ $+ \frac{9 g_2^4}{4}$ $- 9 g_2^2 \lambda_{H2}$ $+ \frac{3 g_Y^4}{4}$ $- 3 g_Y^2 \lambda_{H2}$ $+ 4 \lambda_{12} \lambda_{H1}$ $+ 6 \lambda_H \lambda_{H2}$ $+ 4 \lambda_{H2}^2$ $+ 6 \lambda_{H2} \lambda_{S_2}$ $+ \frac{3 \lambda_{H2}^{\text{Adj}2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H2} (2 \text{Tr} \left(y_2 (y_2)^\dagger \right) + 6 \text{Tr} \left(Y_d (Y_d)^\dagger \right) + 2 \text{Tr} \left(Y_e (Y_e)^\dagger \right) + 6 \text{Tr} \left(Y_u (Y_u)^\dagger \right) + 4 \lambda_{S_2}^{\text{Adj}})$ $- 2 \text{Tr} \left(y_2 (y_2)^\dagger Y_e (Y_e)^\dagger \right)$ $+ 4 \lambda_{T3} (\lambda_{T3})^*$ $+ 6 g_2^4$ $- \frac{33 g_2^2 \lambda_{H2}}{2}$ $+ 3 g_Y^4$ $- \frac{15 g_Y^2 \lambda_{H2}}{2}$ $+ 2 \lambda_{12} \lambda_{H1}$ $+ 6 \lambda_H \lambda_{H2}$ $+ 4 \lambda_{H2}^2$ $+ 8 \lambda_{H2} \lambda_{S_2}$ $+ 6 \lambda_{H2}^{\text{Adj}2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta \lambda_{H2}^{(1)}$
								$\lambda_{H2} (4 \text{Tr} \left(y_2 (y_2)^\dagger \right) + 6 \text{Tr} \left(Y_d (Y_d)^\dagger \right) + 2 \text{Tr} \left(Y_e (Y_e)^\dagger \right) + 6 \text{Tr} \left(Y_u (Y_u)^\dagger \right))$ $- 4 \text{Tr} \left(y_2 (y_2)^\dagger Y_e (Y_e)^\dagger \right)$ $+ 12 \lambda_{T3} (\lambda_{T3})^*$ $- \frac{9 g_2^2 \lambda_{H2}}{2}$ $- \frac{3 g_Y^2 \lambda_{H2}}{2}$ $+ 6 \lambda_{12} \lambda_{H1}$ $+ 6 \lambda_H \lambda_{H2}$ $+ 4 \lambda_{H2}^2$ $+ 4 \lambda_{H2} \lambda_{S_2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{H2} (2 \text{Tr} \left(y_2 (y_2)^\dagger \right) + 6 \text{Tr} \left(Y_d (Y_d)^\dagger \right) + 2 \text{Tr} \left(Y_e (Y_e)^\dagger \right) + 6 \text{Tr} \left(Y_u (Y_u)^\dagger \right) + 4 \lambda_{S_2}^{\text{Adj}})$ $- 2 \text{Tr} \left(y_2 (y_2)^\dagger Y_e (Y_e)^\dagger \right)$ $+ 4 \lambda_{T3} (\lambda_{T3})^*$ $+ 6 g_2^4$ $- \frac{33 g_2^2 \lambda_{H2}}{2}$ $+ 3 g_Y^4$ $- \frac{15 g_Y^2 \lambda_{H2}}{2}$ $+ \lambda_{H1} (6 \lambda_{12} + 4 \lambda_{12}^{\text{Cross}})$ $+ 6 \lambda_H \lambda_{H2}$ $+ 4 \lambda_{H2}^2$ $+ 8 \lambda_{H2} \lambda_{S_2}$ $+ 6 \lambda_{H2}^{\text{Adj}^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	

$$\beta_{\lambda_{12}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{12}}{d\mu}=\beta_{\lambda_{12}}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{12}}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$ \begin{aligned} &\lambda_{12}(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right))\\ &+\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\frac{3}{2}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+9\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*\right)\right)\\ &+\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\left(\frac{3}{2}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+9\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*\right)\right)\\ &+\frac{3}{2}\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*\right)\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)\\ &-2\text{Tr}\left(y_1\left(y_1\right)^\dagger y_2\left(y_2\right)^\dagger\right)\\ &-4\text{Tr}\left(y_1\left(y_2\right)^\dagger y_2\left(y_1\right)^\dagger\right)\\ &+2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\\ &+24\lambda_{S_1^\dagger 2 S_2^\dagger 2}\left(\lambda_{S_1^\dagger 2 S_2^\dagger 2}\right)^*\\ &+3\lambda_{T3}\left(\lambda_{T3}\right)^*\\ &+\frac{9 g_2^4}{4}\\ &-9 g_2^2 \lambda_{12}\\ &+\frac{3 g_Y^4}{4}\\ &-3 g_Y^2 \lambda_{12}\\ &+4 \lambda_{12}^2\\ &+6 \lambda_{12} \lambda_{S_1}\\ &+6 \lambda_{12} \lambda_{S_2}\\ &+\frac{3 \lambda_{12}^{\text{Adj}^2}}{4}\\ &+4 \lambda_{H 1} \lambda_{H 2} \end{aligned} $

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{12}}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{12}(3 \text{Tr}(y_1 (y_1)^\dagger) + 3 \text{Tr}(y_2 (y_2)^\dagger))$ $+ \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left(\frac{9}{4} \text{Tr}(y_1 (y_2)^\dagger) + 9 \left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \right)$ $+ \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\frac{9}{4} \text{Tr}(y_1 (y_2)^\dagger) + 9 \left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \right)$ $+ \frac{9}{4} \left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \text{Tr}(y_2 (y_1)^\dagger)$ $- \frac{9}{2} \text{Tr}(y_1 (y_1)^\dagger y_2 (y_2)^\dagger)$ $- 3 \text{Tr}(y_1 (y_2)^\dagger y_2 (y_1)^\dagger)$ $+ 2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ $+ 24\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$ $+ 3\lambda_{T3} (\lambda_{T3})^*$ $+ \frac{9g_2^4}{4}$ $- 9g_2^2 \lambda_{12}$ $+ \frac{3g_Y^4}{4}$ $- 3g_Y^2 \lambda_{12}$ $+ 4\lambda_{12}^2$ $+ 6\lambda_{12} \lambda_{S_1}$ $+ 6\lambda_{12} \lambda_{S_2}$ $+ \frac{3\lambda_{12}^{\text{Adj}2}}{4}$ $+ 4\lambda_{H1} \lambda_{H2}$
								$\lambda_{12}(4 \text{Tr}(y_1 (y_1)^\dagger) + 2 \text{Tr}(y_2 (y_2)^\dagger) + 4\lambda_{S_2}^{\text{Adj}})$ $- 4 \text{Tr}(y_1 (y_1)^\dagger y_2 (y_2)^\dagger)$ $+ 4\lambda_{T3} (\lambda_{T3})^*$ $- 12g_2^2 \lambda_{12}$ $- 6g_Y^2 \lambda_{12}$ $+ 4\lambda_{12}^2$ $+ 4\lambda_{12} \lambda_{S_1}$ $+ 8\lambda_{12} \lambda_{S_2}$ $+ 4\lambda_{H1} \lambda_{H2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta \lambda_{12}^{(1)}$
								$\lambda_{12}(2 \operatorname{Tr}\left(y_1 (y_1)^\dagger\right) + 4 \operatorname{Tr}\left(y_2 (y_2)^\dagger\right) + 4 \lambda_{S_1}^{\text{Adj}}$ $- 4 \operatorname{Tr}\left(y_1 (y_1)^\dagger y_2 (y_2)^\dagger\right)$ $+ 4 \lambda_{T3} (\lambda_{T3})^*$ $- 12 g_2^2 \lambda_{12}$ $- 6 g_Y^2 \lambda_{12}$ $+ 4 \lambda_{12}^2$ $+ 8 \lambda_{12} \lambda_{S_1}$ $+ 4 \lambda_{12} \lambda_{S_2}$ $+ 4 \lambda_{H1} \lambda_{H2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{12} \left(2 \operatorname{Tr}\left(y_1 (y_1)^\dagger\right) + 2 \operatorname{Tr}\left(y_2 (y_2)^\dagger\right) + \frac{16 \lambda_{12}^{\text{Cross}}}{3} + 4 \lambda_{S_1}^{\text{Adj}} + 4 \lambda_{S_2}^{\text{Adj}} \right)$ $+ \lambda_{12}^{\text{Cross}} \left(\frac{4}{3} \operatorname{Tr}\left(y_1 (y_1)^\dagger\right) + \frac{4}{3} \operatorname{Tr}\left(y_2 (y_2)^\dagger\right) \right)$ $- 2 \operatorname{Tr}\left(y_1 (y_1)^\dagger y_2 (y_2)^\dagger\right)$ $+ \frac{4}{3} \lambda_{T3} (\lambda_{T3})^*$ $+ 16 g_2^4$ $+ g_2^2 (-24 \lambda_{12} - 16 \lambda_{12}^{\text{Cross}})$ $+ g_Y^2 (-6 \lambda_{12} - 4 \lambda_{12}^{\text{Cross}})$ $+ 4 \lambda_{12}^2$ $+ \lambda_{S_1} \left(8 \lambda_{12} + \frac{16 \lambda_{12}^{\text{Cross}}}{3} \right)$ $+ \lambda_{S_2} \left(8 \lambda_{12} + \frac{16 \lambda_{12}^{\text{Cross}}}{3} \right)$ $+ 16 \lambda_{12}^{\text{Adj}^2}$ $+ \frac{32 \lambda_{12}^{\text{Cross}^2}}{3}$ $+ \frac{8 \lambda_{12}^{\text{Cross}} \lambda_{S_1}^{\text{Adj}}}{3}$ $+ \frac{8 \lambda_{12}^{\text{Cross}} \lambda_{S_2}^{\text{Adj}}}{3}$ $+ 4 \lambda_{H1} \lambda_{H2}$

$\beta_{\lambda_{T3}}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{T3}}{d\mu}=\beta_{\lambda_{T3}}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{T3}}^{(1)}$
								$\lambda_{T3}\left(\text{Tr}\left(y_1\ (y_1)^\dagger\right)+\text{Tr}\left(y_2\ (y_2)^\dagger\right)+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$ $+2\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$ $+2\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\text{Tr}\left(y_1\ (y_2)^\dagger\right)$ $-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ $+4\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ $+4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$ $-9g_2^2\lambda_{T3}$ $-3g_Y^2\lambda_{T3}$ $+2\lambda_{12}\lambda_{T3}$ $+2\lambda_H\lambda_{T3}$ $+4\lambda_{H1}\lambda_{T3}$ $+4\lambda_{H2}\lambda_{T3}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	
								$\lambda_{T3}\left(\frac{3}{2}\text{Tr}\left(y_1\ (y_1)^\dagger\right)+\frac{3}{2}\text{Tr}\left(y_2\ (y_2)^\dagger\right)+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$ $+3\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$ $+3\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\text{Tr}\left(y_1\ (y_2)^\dagger\right)$ $-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ $+4\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ $+4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$ $-9g_2^2\lambda_{T3}$ $-3g_Y^2\lambda_{T3}$ $+2\lambda_{12}\lambda_{T3}$ $+2\lambda_H\lambda_{T3}$ $+4\lambda_{H1}\lambda_{T3}$ $+4\lambda_{H2}\lambda_{T3}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	
								$\lambda_{T3}(2\text{Tr}\left(y_1\ (y_1)^\dagger\right)+\text{Tr}\left(y_2\ (y_2)^\dagger\right)+6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)+4\lambda_{H2}^{\text{Adj}})$ $-\frac{21g_2^2\lambda_{T3}}{2}$ $-\frac{9g_Y^2\lambda_{T3}}{2}$ $+2\lambda_{12}\lambda_{T3}$ $+2\lambda_H\lambda_{T3}$ $+4\lambda_{H1}\lambda_{T3}$ $+4\lambda_{H2}\lambda_{T3}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{T3}}^{(1)}$
								$\lambda_{T3}(\text{Tr}\left(y_1\ (y_1)^\dagger\right) + 2\text{Tr}\left(y_2\ (y_2)^\dagger\right) + 6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) - 4\lambda_{H1}^{\text{Adj}})$ $- \frac{21g_2^2\lambda_{T3}}{2}$ $- \frac{9g_Y^2\lambda_{T3}}{2}$ $+ 2\lambda_{12}\lambda_{T3}$ $+ 2\lambda_H\lambda_{T3}$ $+ 4\lambda_{H1}\lambda_{T3}$ $+ 4\lambda_{H2}\lambda_{T3}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{T3}\left(\text{Tr}\left(y_1\ (y_1)^\dagger\right) + \text{Tr}\left(y_2\ (y_2)^\dagger\right) + 6\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\text{Tr}\left(Y_u\ (Y_u)^\dagger\right) - \frac{4\lambda_{12}^{\text{Adj}}}{3} - 2\lambda_{12}^{\text{Cross}} - 2\lambda_{H1}^{\text{Adj}} - \frac{4\lambda_{H2}^{\text{Adj}}}{3}\right)$ $- \frac{33g_2^2\lambda_{T3}}{2}$ $- \frac{9g_Y^2\lambda_{T3}}{2}$ $+ 2\lambda_{12}\lambda_{T3}$ $+ 2\lambda_H\lambda_{T3}$ $+ 4\lambda_{H1}\lambda_{T3}$ $+ 4\lambda_{H2}\lambda_{T3}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	

$$\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{12}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{12}^{\text{Adj}}(2\text{Tr}\left(y_1\ (y_1)^\dagger\right)+2\text{Tr}\left(y_2\ (y_2)^\dagger\right))$ $+\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\frac{2}{3}\text{Tr}\left(y_1\ (y_2)^\dagger\right)+\frac{4}{3}(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*)\right)$ $+\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\left(\frac{2}{3}\text{Tr}\left(y_1\ (y_2)^\dagger\right)+\frac{4}{3}((\lambda_{S_1^\dagger S_2 S_2^\dagger 2})^*+5\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*)\right)$ $+\frac{2}{3}((\lambda_{S_1^\dagger S_2 S_2^\dagger 2})^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*)\text{Tr}\left(y_2\ (y_1)^\dagger\right)$ $-\frac{8}{3}\text{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$ $+\frac{64}{3}\lambda_{S_1^\dagger 2 S_2^\dagger 2}\left(\lambda_{S_1^\dagger 2 S_2^\dagger 2}\right)^*$ $+\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$ $+g_Y{}^2\left(-6g_2{}^2-3\lambda_{12}^{\text{Adj}}\right)$ $-9g_2{}^2\lambda_{12}^{\text{Adj}}$ $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
								$\lambda_{12}^{\text{Adj}}(3\text{Tr}\left(y_1\ (y_1)^\dagger\right)+3\text{Tr}\left(y_2\ (y_2)^\dagger\right))$ $+\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\text{Tr}\left(y_1\ (y_2)^\dagger\right)+\frac{4}{3}(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*)\right)$ $+\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\left(\text{Tr}\left(y_1\ (y_2)^\dagger\right)+\frac{4}{3}((\lambda_{S_1^\dagger S_2 S_2^\dagger 2})^*+5\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*)\right)$ $+((\lambda_{S_1^\dagger S_2 S_2^\dagger 2})^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*)\text{Tr}\left(y_2\ (y_1)^\dagger\right)$ $-\frac{2}{3}\text{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$ $-\frac{8}{3}\text{Tr}\left(y_1\ (y_2)^\dagger\ y_2\ (y_1)^\dagger\right)$ $+\frac{64}{3}\lambda_{S_1^\dagger 2 S_2^\dagger 2}\left(\lambda_{S_1^\dagger 2 S_2^\dagger 2}\right)^*$ $+\frac{4}{3}\lambda_{T3}\left(\lambda_{T3}\right)^*$ $+g_Y{}^2\left(-6g_2{}^2-3\lambda_{12}^{\text{Adj}}\right)$ $-9g_2{}^2\lambda_{12}^{\text{Adj}}$ $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
								$\lambda_{12}^{\text{Adj}} \left(2 \text{Tr} \left(y_1 (y_1)^\dagger \right) + 2 \text{Tr} \left(y_2 (y_2)^\dagger \right) + \frac{16 \lambda_{12}^{\text{Cross}}}{3} \right)$ $+ \frac{2}{3} \text{Tr} \left(y_1 (y_1)^\dagger y_2 (y_2)^\dagger \right)$ $- \frac{2}{9} \lambda_{T3} (\lambda_{T3})^*$ $- 24 g_2^2 \lambda_{12}^{\text{Adj}}$ $- 6 g_Y^2 \lambda_{12}^{\text{Adj}}$ $+ 8 \lambda_{12} \lambda_{12}^{\text{Adj}}$ $+ 2 \lambda_{12}^{\text{Adj}} \lambda_{S_1}$ $+ 16 \lambda_{12}^{\text{Adj}} \lambda_{S_1}^{\text{Adj}}$ $+ 2 \lambda_{12}^{\text{Adj}} \lambda_{S_2}$ $+ 16 \lambda_{12}^{\text{Adj}} \lambda_{S_2}^{\text{Adj}}$ $+ 3 \lambda_{H1}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	

$\beta_{\lambda_{12}^{\text{Cross}}}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{12}^{\text{Cross}}}{d\mu}=\beta_{\lambda_{12}^{\text{Cross}}}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{12}^{\text{Cross}}}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
$\lambda_{12}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+8\lambda_{12}^{\text{Cross}}+\lambda_{S_1}^{\text{Adj}}+\lambda_{S_2}^{\text{Adj}}\right)$ $+\lambda_{12}^{\text{Cross}}\left(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\right)$ $-\frac{7}{6}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$ $-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^*$ $+\frac{33g_2^4}{4}$ $+g_2^2\left(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}}\right)$ $+g_Y^2\left(-\frac{3\lambda_{12}}{2}-6\lambda_{12}^{\text{Cross}}\right)$ $+\lambda_{12}^2$ $+\lambda_{S_1}\left(2\lambda_{12}+3\lambda_{12}^{\text{Cross}}\right)$ $+\lambda_{S_2}\left(2\lambda_{12}+3\lambda_{12}^{\text{Cross}}\right)$ $+4\lambda_{12}^{\text{Adj}^2}$ $+11\lambda_{12}^{\text{Cross}^2}$ $-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$ $-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$ $+\lambda_{H1}\lambda_{H2}$								
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	

$$\beta_{\lambda_{H1}^{\text{Adj}}}^{(1)}$$

$$16\pi^2 \mu \frac{d\lambda_{H1}^{\text{Adj}}}{d\mu} = \beta_{\lambda_{H1}^{\text{Adj}}}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{H1}^{\text{Adj}}}^{(1)}$
								$\lambda_{H1}^{\text{Adj}} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger))$ $+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(-\frac{2}{3} \text{Tr} (y_1 (y_2)^\dagger) - \frac{4}{3} (2 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + \left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \right)^*) \right)$ $- \frac{2}{3} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr} (y_2 (y_1)^\dagger)$ $+ \frac{8}{3} \text{Tr} (y_1 (y_1)^\dagger Y_e (Y_e)^\dagger)$ $+ \frac{64}{3} \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$ $- \frac{4}{3} \lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ $+ \frac{16}{3} \lambda_{T3} (\lambda_{T3})^*$ $+ g_Y^2 \left(-6 g_2^2 - 3 \lambda_{H1}^{\text{Adj}} \right)$ $- 9 g_2^2 \lambda_{H1}^{\text{Adj}}$ $+ \lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ $+ 2 \lambda_H \lambda_{H1}^{\text{Adj}}$ $+ 8 \lambda_{H1} \lambda_{H1}^{\text{Adj}}$ $+ 2 \lambda_{H1}^{\text{Adj}} \lambda_{S_1}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H1}^{\text{Adj}} (3 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger))$ $+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(-\text{Tr} (y_1 (y_2)^\dagger) - \frac{4}{3} (2 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + \left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \right)^*) \right)$ $- \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr} (y_2 (y_1)^\dagger)$ $- \frac{4}{3} \text{Tr} (y_1 (y_1)^\dagger Y_e (Y_e)^\dagger)$ $+ \frac{64}{3} \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$ $- \frac{4}{3} \lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ $+ \frac{16}{3} \lambda_{T3} (\lambda_{T3})^*$ $+ g_Y^2 \left(-6 g_2^2 - 3 \lambda_{H1}^{\text{Adj}} \right)$ $- 9 g_2^2 \lambda_{H1}^{\text{Adj}}$ $+ \lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ $+ 2 \lambda_H \lambda_{H1}^{\text{Adj}}$ $+ 8 \lambda_{H1} \lambda_{H1}^{\text{Adj}}$ $+ 2 \lambda_{H1}^{\text{Adj}} \lambda_{S_1}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H1}^{\text{Adj}} (3 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger))$ $+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(-\text{Tr} (y_1 (y_2)^\dagger) - \frac{4}{3} (2 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + \left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \right)^*) \right)$ $- \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr} (y_2 (y_1)^\dagger)$ $- \frac{4}{3} \text{Tr} (y_1 (y_1)^\dagger Y_e (Y_e)^\dagger)$ $+ \frac{64}{3} \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$ $- \frac{4}{3} \lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ $+ \frac{16}{3} \lambda_{T3} (\lambda_{T3})^*$ $+ g_Y^2 \left(-6 g_2^2 - 3 \lambda_{H1}^{\text{Adj}} \right)$ $- 9 g_2^2 \lambda_{H1}^{\text{Adj}}$ $+ \lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ $+ 2 \lambda_H \lambda_{H1}^{\text{Adj}}$ $+ 8 \lambda_{H1} \lambda_{H1}^{\text{Adj}}$ $+ 2 \lambda_{H1}^{\text{Adj}} \lambda_{S_1}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta \lambda_{H1}^{(1) \text{Adj}}$
								$\lambda_{H1}^{\text{Adj}} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 16 \lambda_{S_1}^{\text{Adj}})$ $- \frac{4}{3} \text{Tr} (y_1 (y_1)^\dagger Y_e (Y_e)^\dagger)$ $- \frac{8}{3} \lambda_{T3} (\lambda_{T3})^*$ $+ g_Y^2 \left(-4 g_2^2 - \frac{15 \lambda_{H1}^{\text{Adj}}}{2} \right)$ $- \frac{33 g_2^2 \lambda_{H1}^{\text{Adj}}}{2}$ $+ 2 \lambda_H \lambda_{H1}^{\text{Adj}}$ $+ 8 \lambda_{H1} \lambda_{H1}^{\text{Adj}}$ $+ 2 \lambda_{H1}^{\text{Adj}} \lambda_{S_1}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{H1}^{\text{Adj}} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 16 \lambda_{S_1}^{\text{Adj}})$ $- \frac{4}{3} \text{Tr} (y_1 (y_1)^\dagger Y_e (Y_e)^\dagger)$ $- \frac{4}{3} \lambda_{T3} (\lambda_{T3})^*$ $- \frac{33 g_2^2 \lambda_{H1}^{\text{Adj}}}{2}$ $- \frac{3 g_Y^2 \lambda_{H1}^{\text{Adj}}}{2}$ $+ 12 \lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$ $+ 2 \lambda_H \lambda_{H1}^{\text{Adj}}$ $+ 8 \lambda_{H1} \lambda_{H1}^{\text{Adj}}$ $+ 2 \lambda_{H1}^{\text{Adj}} \lambda_{S_1}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	

$$\beta_{\lambda_{H2}^{\text{Adj}}}^{(1)}$$

$$16\pi^2 \mu \frac{d\lambda_{H2}^{\text{Adj}}}{d\mu} = \beta_{\lambda_{H2}^{\text{Adj}}}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{H2}^{\text{Adj}}}^{(1)}$
								$\lambda_{H2}^{\text{Adj}} (2 \text{Tr} (y_2 (y_2)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger))$ $+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (-2 \text{Tr} (y_1 (y_2)^\dagger) - 4(2 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + \left(\lambda_{S_1^\dagger S_2 S_2^\dagger} \right)^*))$ $- 2 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr} (y_2 (y_1)^\dagger)$ $+ 8 \text{Tr} (y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$ $- 4 \lambda_{S_1^\dagger S_2 S_2^\dagger} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ $+ 64 \lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$ $+ 16 \lambda_{T3} (\lambda_{T3})^*$ $+ g_Y^2 \left(6 g_2^2 - 3 \lambda_{H2}^{\text{Adj}} \right)$ $- 9 g_2^2 \lambda_{H2}^{\text{Adj}}$ $+ \lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$ $+ 2 \lambda_H \lambda_{H2}^{\text{Adj}}$ $+ 8 \lambda_{H2} \lambda_{H2}^{\text{Adj}}$ $+ 2 \lambda_{H2}^{\text{Adj}} \lambda_{S_2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H2}^{\text{Adj}} (3 \text{Tr} (y_2 (y_2)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger))$ $+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (-3 \text{Tr} (y_1 (y_2)^\dagger) - 4(2 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + \left(\lambda_{S_1^\dagger S_2 S_2^\dagger} \right)^*))$ $- 3 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr} (y_2 (y_1)^\dagger)$ $- 4 \text{Tr} (y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$ $- 4 \lambda_{S_1^\dagger S_2 S_2^\dagger} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ $+ 64 \lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$ $+ 16 \lambda_{T3} (\lambda_{T3})^*$ $+ g_Y^2 \left(6 g_2^2 - 3 \lambda_{H2}^{\text{Adj}} \right)$ $- 9 g_2^2 \lambda_{H2}^{\text{Adj}}$ $+ \lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$ $+ 2 \lambda_H \lambda_{H2}^{\text{Adj}}$ $+ 8 \lambda_{H2} \lambda_{H2}^{\text{Adj}}$ $+ 2 \lambda_{H2}^{\text{Adj}} \lambda_{S_2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H2}^{\text{Adj}} (3 \text{Tr} (y_2 (y_2)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger))$ $+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (-3 \text{Tr} (y_1 (y_2)^\dagger) - 4(2 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + \left(\lambda_{S_1^\dagger S_2 S_2^\dagger} \right)^*))$ $- 3 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr} (y_2 (y_1)^\dagger)$ $- 4 \text{Tr} (y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$ $- 4 \lambda_{S_1^\dagger S_2 S_2^\dagger} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ $+ 64 \lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$ $+ 16 \lambda_{T3} (\lambda_{T3})^*$ $+ g_Y^2 \left(6 g_2^2 - 3 \lambda_{H2}^{\text{Adj}} \right)$ $- 9 g_2^2 \lambda_{H2}^{\text{Adj}}$ $+ \lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$ $+ 2 \lambda_H \lambda_{H2}^{\text{Adj}}$ $+ 8 \lambda_{H2} \lambda_{H2}^{\text{Adj}}$ $+ 2 \lambda_{H2}^{\text{Adj}} \lambda_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta \lambda_{H2}^{Adj(1)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{H2}^{Adj} (2 \text{Tr} (y_2 (y_2)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 16 \lambda_{S_2}^{Adj})$ $+ \frac{4}{3} \text{Tr} (y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$ $+ \frac{8}{3} \lambda_{T3} (\lambda_{T3})^*$ $+ g_Y^2 \left(4 g_2^2 - \frac{15 \lambda_{H2}^{Adj}}{2} \right)$ $- \frac{33 g_2^2 \lambda_{H2}^{Adj}}{2}$ $+ 2 \lambda_H \lambda_{H2}^{Adj}$ $+ 8 \lambda_{H2} \lambda_{H2}^{Adj}$ $+ 2 \lambda_{H2}^{Adj} \lambda_{S_2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{H2}^{Adj} (2 \text{Tr} (y_2 (y_2)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 16 \lambda_{S_2}^{Adj})$ $+ \frac{4}{3} \text{Tr} (y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$ $- \frac{8}{9} \lambda_{T3} (\lambda_{T3})^*$ $+ g_Y^2 \left(4 g_2^2 - \frac{15 \lambda_{H2}^{Adj}}{2} \right)$ $- \frac{33 g_2^2 \lambda_{H2}^{Adj}}{2}$ $+ 12 \lambda_{12}^{Adj} \lambda_{H1}^{Adj}$ $+ 2 \lambda_H \lambda_{H2}^{Adj}$ $+ 8 \lambda_{H2} \lambda_{H2}^{Adj}$ $+ 2 \lambda_{H2}^{Adj} \lambda_{S_2}$

$$\beta_{\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{d\mu}=\beta_{\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}^{(1)}$
								$\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}(2\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)+6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$ $+(\lambda_{T3})^*\text{Tr}\Big(y_2\ (y_1)^\dagger\Big)$ $-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_{T3})^*$ $+2\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}(\lambda_{T3})^*$ $-9g_2^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ $-3g_Y^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ $+2\lambda_H\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ $+8\lambda_{H1}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ $+\frac{4\lambda_{H1}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$ $+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\lambda_{S1}$ $+4\lambda_{H^\dagger H^\dagger S_2 S_2}\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}(3\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)+6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$ $+\frac{3}{2}(\lambda_{T3})^*\text{Tr}\Big(y_2\ (y_1)^\dagger\Big)$ $-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_{T3})^*$ $+2\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}(\lambda_{T3})^*$ $-9g_2^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ $-3g_Y^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ $+2\lambda_H\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ $+8\lambda_{H1}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ $+\frac{4\lambda_{H1}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$ $+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\lambda_{S1}$ $+4\lambda_{H^\dagger H^\dagger S_2 S_2}\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}(3\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)+6\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)+2\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)+6\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$ $+\frac{3}{2}(\lambda_{T3})^*\text{Tr}\Big(y_2\ (y_1)^\dagger\Big)$ $-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_{T3})^*$ $+2\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}(\lambda_{T3})^*$ $-9g_2^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ $-3g_Y^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ $+2\lambda_H\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ $+8\lambda_{H1}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ $+\frac{4\lambda_{H1}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$ $+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\lambda_{S1}$ $+4\lambda_{H^\dagger H^\dagger S_2 S_2}\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$$\beta_{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{d\mu}=\beta_{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}^{(1)}$
								$\lambda_{H1}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)$ $+\lambda_{H1}^{\text{Adj}}\left(-\frac{1}{8}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)-\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}-\frac{\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}}{4}\right)$ $+\lambda_{H2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)$ $+\lambda_{H2}^{\text{Adj}}\left(-\frac{3}{8}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)-\frac{3\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}-\frac{3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}}{4}\right)$ $+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-4\text{Tr}\left(y_2\left(y_1\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$ $+6\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ $-6\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\left(\lambda_{T3}\right)^*$ $-9g_2^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ $-3g_Y^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ $+2\lambda_{12}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ $+3\lambda_H\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ $-6\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\lambda_{T3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{(1)\text{Cross}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$ \begin{aligned} & \lambda_{H1} \left(\frac{3}{4} \text{Tr} \left(y_2 (y_1)^\dagger \right) + 4 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + 3 \lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \right) \\ & + \lambda_{H1}^{\text{Adj}} \left(-\frac{3}{16} \text{Tr} \left(y_2 (y_1)^\dagger \right) - \frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}}{4} \right) \\ & + \lambda_{H2} \left(\frac{3}{4} \text{Tr} \left(y_2 (y_1)^\dagger \right) + 4 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} + 3 \lambda_{S_1^\dagger S_2 S_2^{\dagger 2}} \right) \\ & + \lambda_{H2}^{\text{Adj}} \left(-\frac{9}{16} \text{Tr} \left(y_2 (y_1)^\dagger \right) - \frac{3 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{3 \lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}}{4} \right) \\ & + \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\frac{3}{2} \text{Tr} \left(y_1 (y_1)^\dagger \right) + \frac{3}{2} \text{Tr} \left(y_2 (y_2)^\dagger \right) + 6 \text{Tr} \left(Y_d (Y_d)^\dagger \right) + 2 \text{Tr} \left(Y_e (Y_e)^\dagger \right) + 6 \text{Tr} \left(Y_u (Y_u)^\dagger \right) \right) \\ & + 6 \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \\ & - 6 \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* (\lambda_{T3})^* \\ & - 9 g_2^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \\ & - 3 g_Y^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \\ & + 2 \lambda_{12} \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \\ & + 3 \lambda_H \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \\ & - 6 \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \lambda_{T3} \end{aligned} $
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$$\beta_{\lambda_{H^\dagger H^\dagger S_2 S_2}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{H^\dagger H^\dagger S_2 S_2}}{d\mu}=\beta_{\lambda_{H^\dagger H^\dagger S_2 S_2}}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{H^\dagger H^\dagger S_2 S_2}}^{(1)}$
								$\lambda_{H^\dagger H^\dagger S_2 S_2}(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$ $+(\lambda_{T3})^*\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)$ $+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$ $+2\left(\lambda_{T3}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*-\left(\lambda_{H H^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{H^\dagger H^\dagger S_2 S_2}$ $-3g_Y^2\lambda_{H^\dagger H^\dagger S_2 S_2}$ $+2\lambda_H\lambda_{H^\dagger H^\dagger S_2 S_2}$ $+8\lambda_{H2}\lambda_{H^\dagger H^\dagger S_2 S_2}$ $+4\lambda_{H2}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_2 S_2}$ $+2\lambda_{H^\dagger H^\dagger S_2 S_2}\lambda_{S_2}$
								$\lambda_{H^\dagger H^\dagger S_2 S_2}(3\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$ $+\frac{3}{2}\left(\lambda_{T3}\right)^*\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)$ $+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$ $+2\left(\lambda_{T3}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*-\left(\lambda_{H H^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{H^\dagger H^\dagger S_2 S_2}$ $-3g_Y^2\lambda_{H^\dagger H^\dagger S_2 S_2}$ $+2\lambda_H\lambda_{H^\dagger H^\dagger S_2 S_2}$ $+8\lambda_{H2}\lambda_{H^\dagger H^\dagger S_2 S_2}$ $+4\lambda_{H2}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_2 S_2}$ $+2\lambda_{H^\dagger H^\dagger S_2 S_2}\lambda_{S_2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$$\beta_{\lambda_{S_1}^{\text{Adj}}}^{(1)}$$

$$16\pi^2\,\mu\frac{d\lambda_{S_1}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{S_1}^{\text{Adj}}}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{S_1}^{\text{Adj}}}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
								$\lambda_{S_1}\left(\frac{1}{2}\,\text{Tr}\Big(y_1\,(y_1)^\dagger\Big)+13\lambda_{S_1}^{\text{Adj}}\right)$ $+4\lambda_{S_1}^{\text{Adj}}\,\text{Tr}\Big(y_1\,(y_1)^\dagger\Big)$ $-\text{Tr}\Big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\Big)$ $+\frac{3g_2^4}{4}$ $+g_Y^2\left(9g_2^2-\frac{3\lambda_{S_1}}{2}-12\lambda_{S_1}^{\text{Adj}}\right)$ $+g_2^2(-3\lambda_{S_1}-24\lambda_{S_1}^{\text{Adj}})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{3g_Y^4}{2}$ $+\frac{\lambda_{12}^2}{4}$ $+\frac{\lambda_{H1}^2}{2}$ $+3\lambda_{H1}^{\text{Adj}2}$ $+\frac{7\lambda_{S_1}^2}{4}$ $+22\lambda_{S_1}^{\text{Adj}2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{S_1}^{\text{Adj}}}^{(1)}$
								$\lambda_{S_1} \left(\frac{1}{2} \text{Tr} \left(y_1 (y_1)^\dagger \right) + 13 \lambda_{S_1}^{\text{Adj}} \right)$ $+ 4 \lambda_{S_1}^{\text{Adj}} \text{Tr} \left(y_1 (y_1)^\dagger \right)$ $- \text{Tr} \left(y_1 (y_1)^\dagger y_1 (y_1)^\dagger \right)$ $+ \frac{3 g_2^4}{4}$ $+ g_2^2 (-3 \lambda_{S_1} - 24 \lambda_{S_1}^{\text{Adj}})$ $+ \frac{3 \lambda_{12}^2}{4}$ $+ \lambda_{12} \lambda_{12}^{\text{Cross}}$ $+ 12 \lambda_{12}^{\text{Adj}^2}$ $- \frac{\lambda_{12}^{\text{Cross}^2}}{2}$ $+ \frac{\lambda_{H1}^2}{2}$ $+ 3 \lambda_{H1}^{\text{Adj}^2}$ $+ \frac{7 \lambda_{S_1}^2}{4}$ $+ 22 \lambda_{S_1}^{\text{Adj}^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	

$$\beta_{\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}}{d\mu}=\beta_{\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\begin{aligned} & \lambda_{12}(\text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)+6 \lambda_{S_1^{\dagger} S_2 S_2^{\dagger} 2}+6 \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}) \\ & +\lambda_{12}^{\text {Adj }}\left(\frac{1}{12} \text {Tr}\left(y_2\left(y_1\right)^{\dagger}\right)+\frac{\lambda_{S_1^{\dagger} S_2 S_2^{\dagger} 2}}{6}+\frac{5 \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}}{6}\right) \\ & +\lambda_{S_1}(\text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)+12 \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}) \\ & +\lambda_{S_1^{\dagger 2} S_2^{\dagger} 2}(2 \text {Tr}\left(y_1\left(y_2\right)^{\dagger}\right)+4\left(\left(\lambda_{S_1^{\dagger} S_2 S_2^{\dagger} 2}\right)^{*}+5\left(\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}\right)^{*}\right)) \\ & +\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}(3 \text {Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\text {Tr}\left(y_2\left(y_2\right)^{\dagger}\right)) \\ & -4 \text {Tr}\left(y_1\left(y_1\right)^{\dagger} y_2\left(y_1\right)^{\dagger}\right) \\ & -9 g_2^2 \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \\ & -3 g_Y^2 \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \\ & +2 \lambda_{H 1} \lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text {Cross }} \\ & -\frac{\lambda_{H 1}^{\text {Adj }} \lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text {Cross }}}{6} \\ & +4 \lambda_{H^{\dagger} H^{\dagger} S_1^{\dagger} S_1^{\dagger}} \lambda_{T 3} \end{aligned}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\begin{aligned} & \lambda_{12}\left(\frac{3}{2} \text {Tr}\left(y_2\left(y_1\right)^{\dagger}\right)+6 \lambda_{S_1^{\dagger} S_2 S_2^{\dagger} 2}+6 \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}\right) \\ & +\lambda_{12}^{\text {Adj }}\left(\frac{1}{8} \text {Tr}\left(y_2\left(y_1\right)^{\dagger}\right)+\frac{\lambda_{S_1^{\dagger} S_2 S_2^{\dagger} 2}}{6}+\frac{5 \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}}{6}\right) \\ & +\lambda_{S_1}\left(\frac{3}{2} \text {Tr}\left(y_2\left(y_1\right)^{\dagger}\right)+12 \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}\right) \\ & +\lambda_{S_1^{\dagger 2} S_2^{\dagger} 2}(3 \text {Tr}\left(y_1\left(y_2\right)^{\dagger}\right)+4\left(\left(\lambda_{S_1^{\dagger} S_2 S_2^{\dagger} 2}\right)^{*}+5\left(\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}\right)^{*}\right)) \\ & +\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}\left(\frac{9}{2} \text {Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\frac{3}{2} \text {Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\right) \\ & -5 \text {Tr}\left(y_1\left(y_1\right)^{\dagger} y_2\left(y_1\right)^{\dagger}\right) \\ & -9 g_2^2 \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \\ & -3 g_Y^2 \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \\ & +2 \lambda_{H 1} \lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text {Cross }} \\ & -\frac{\lambda_{H 1}^{\text {Adj }} \lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text {Cross }}}{6} \\ & +4 \lambda_{H^{\dagger} H^{\dagger} S_1^{\dagger} S_1^{\dagger}} \lambda_{T 3} \end{aligned}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda}^{(1)}$ $S_1 S_1^\dagger S_2^\dagger$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$$\beta_{\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}{d\mu}=\beta_{\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}^{(1)}$
								$\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}(2\text{Tr}\Big(y_1\ (y_1)^{\dagger}\Big)+2\text{Tr}\Big(y_2\ (y_2)^{\dagger}\Big))$ $+\lambda_{S_1^{\dagger} S_2 S_2^{\dagger 2}}(\text{Tr}\Big(y_2\ (y_1)^{\dagger}\Big)+2\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}})$ $+\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}\text{Tr}\Big(y_2\ (y_1)^{\dagger}\Big)$ $-2\text{Tr}\Big(y_2\ (y_1)^{\dagger}\ y_2\ (y_1)^{\dagger}\Big)$ $+4\lambda_{H^{\dagger} H^{\dagger} S_1^{\dagger} S_1^{\dagger}}\left(\lambda_{H^{\dagger} H^{\dagger} S_2 S_2}\right)^*$ $-9g_2^2\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$ $-3g_Y^2\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$ $+8\lambda_{12}\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$ $+\frac{4\lambda_{12}^{\text{Adj}}\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}{3}$ $+\lambda_{HH^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text{Cross}}{}^2$ $+2\lambda_{S_1}\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$ $+2\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\lambda_{S_2}$ $+5\lambda_{S_1^{\dagger} S_2 S_2^{\dagger 2}}{}^2$ $+5\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}{}^2$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}^{(1)}$
								$\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} (3 \text{Tr} (y_1 (y_1)^{\dagger}) + 3 \text{Tr} (y_2 (y_2)^{\dagger}))$ $+ \lambda_{S_1^{\dagger} S_2 S_2^{\dagger 2}} \left(\frac{3}{2} \text{Tr} (y_2 (y_1)^{\dagger}) + 2 \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \right)$ $+ \frac{3}{2} \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \text{Tr} (y_2 (y_1)^{\dagger})$ $- \frac{5}{2} \text{Tr} (y_2 (y_1)^{\dagger} y_2 (y_1)^{\dagger})$ $+ 4 \lambda_{H^{\dagger} H^{\dagger} S_1^{\dagger} S_1^{\dagger}} \left(\lambda_{H^{\dagger} H^{\dagger} S_2 S_2} \right)^*$ $- 9 g_2^2 \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$ $- 3 g_Y^2 \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$ $+ 8 \lambda_{12} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$ $+ \frac{4 \lambda_{12}^{\text{Adj}} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}{3}$ $+ \lambda_{H H^{\dagger} S_1^{\dagger} S_2^{\dagger}}^{\text{Cross}^2}$ $+ 2 \lambda_{S_1} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$ $+ 2 \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \lambda_{S_2}$ $+ 5 \lambda_{S_1^{\dagger} S_2 S_2^{\dagger 2}}^2$ $+ 5 \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger 2}}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$$\beta_{\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}^{(1)}$$

$$_{16\pi^2\mu}\frac{d\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{d\mu}=\beta_{\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}^{(1)}$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\begin{aligned} & \lambda_{12}(\text{Tr}\left(y_2\ (y_1)^\dagger\right)+6\lambda_{S_1^\dagger S_2 S_2^\dagger 2}+6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}) \\ & +\lambda_{12}^{\text{Adj}}\left(\frac{1}{12}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+\frac{5\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{6}+\frac{\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}}{6}\right) \\ & +\lambda_{S_1^\dagger 2 S_2^\dagger 2}(2\text{Tr}\left(y_1\ (y_2)^\dagger\right)+4(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*)) \\ & +\lambda_{S_2}(\text{Tr}\left(y_2\ (y_1)^\dagger\right)+12\lambda_{S_1^\dagger S_2 S_2^\dagger 2}) \\ & +\lambda_{S_1^\dagger S_2 S_2^\dagger 2}(\text{Tr}\left(y_1\ (y_1)^\dagger\right)+3\text{Tr}\left(y_2\ (y_2)^\dagger\right)) \\ & -4\text{Tr}\left(y_2\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right) \\ & +4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*(\lambda_{T3})^* \\ & -9g_2^2\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \\ & -3g_Y^2\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \\ & +2\lambda_{H2}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \\ & -\frac{\lambda_{H2}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} \end{aligned}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\begin{aligned} & \lambda_{12}\left(\frac{3}{2}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+6\lambda_{S_1^\dagger S_2 S_2^\dagger 2}+6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right) \\ & +\lambda_{12}^{\text{Adj}}\left(\frac{1}{8}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+\frac{5\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{6}+\frac{\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}}{6}\right) \\ & +\lambda_{S_1^\dagger 2 S_2^\dagger 2}(3\text{Tr}\left(y_1\ (y_2)^\dagger\right)+4(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*)) \\ & +\lambda_{S_2}\left(\frac{3}{2}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+12\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right) \\ & +\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\frac{3}{2}\text{Tr}\left(y_1\ (y_1)^\dagger\right)+\frac{9}{2}\text{Tr}\left(y_2\ (y_2)^\dagger\right)\right) \\ & -5\text{Tr}\left(y_2\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right) \\ & +4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*(\lambda_{T3})^* \\ & -9g_2^2\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \\ & -3g_Y^2\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \\ & +2\lambda_{H2}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \\ & -\frac{\lambda_{H2}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} \end{aligned}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_\lambda^{(1)}$
								$S_1^\dagger S_2 S_2^{\dagger 2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—

$\beta_{\lambda_{S_2}^{\text{Adj}}}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_2}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{S_2}^{\text{Adj}}}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta_{\lambda_{S_2}^{\text{Adj}}}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
$\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+13\lambda_{S_2}^{\text{Adj}}\right)$ $+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$ $-\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ $+\frac{3g_2^4}{4}$ $+g_Y^2\left(9g_2^2-\frac{3\lambda_{S_2}}{2}-12\lambda_{S_2}^{\text{Adj}}\right)$ $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$								
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3g_Y^4}{2}$ $+\frac{\lambda_{12}^2}{4}$ $+\frac{\lambda_{H2}^2}{2}$ $+3\lambda_{H2}^{\text{Adj}2}$ $+\frac{7\lambda_{S_2}^2}{4}$ $+22\lambda_{S_2}^{\text{Adj}2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$\beta \lambda_{S_2}^{(1)\text{Adj}}$
								$ \begin{aligned} &\lambda_{S_2} \left(\frac{1}{2} \text{Tr} \left(y_2 (y_2)^\dagger \right) + 13 \lambda_{S_2}^{\text{Adj}} \right) \\ &\quad + 4 \lambda_{S_2}^{\text{Adj}} \text{Tr} \left(y_2 (y_2)^\dagger \right) \\ &\quad - \text{Tr} \left(y_2 (y_2)^\dagger y_2 (y_2)^\dagger \right) \\ &\quad + \frac{3 g_2^4}{4} \\ &\quad + g_Y^2 \left(9 g_2^2 - \frac{3 \lambda_{S_2}}{2} - 12 \lambda_{S_2}^{\text{Adj}} \right) \\ &\quad + g_2^2 (-3 \lambda_{S_2} - 24 \lambda_{S_2}^{\text{Adj}}) \\ &\quad + \frac{3 g_Y^4}{2} \\ &\quad + \frac{3 \lambda_{12}^2}{4} \\ &\quad + \lambda_{12} \lambda_{12}^{\text{Cross}} \\ &\quad + 12 \lambda_{12}^{\text{Adj}^2} \\ &\quad - \frac{\lambda_{12}^{\text{Cross}^2}}{2} \\ &\quad + \frac{\lambda_{H2}^2}{2} \\ &\quad + 3 \lambda_{H2}^{\text{Adj}^2} \\ &\quad + \frac{7 \lambda_{S_2}^2}{4} \\ &\quad + 22 \lambda_{S_2}^{\text{Adj}^2} \end{aligned} $
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	